Logistics Costing and Pricing

Course 4 — Logistics Costing, Risk and Cross-Border Operations · Module 10 · Teaching content for the Theory Builder

Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.

Lesson 10.1 — Components of Total Logistics Cost

Summary

This lesson examines the different expenses that contribute to the total cost of logistics operations. Learners will explore transportation, warehousing, inventory holding, cargo handling, administration, insurance, and distribution costs. The lesson also explains how direct and indirect costs influence operational decisions, pricing, profitability, and customer service.

Learning objectives

- Define total logistics cost and explain its importance in business operations.
- Identify the major components of logistics expenditure.
- Distinguish between direct and indirect logistics costs.
- Explain the differences between fixed and variable costs.
- Describe how transportation, storage, and inventory decisions affect total costs.
- Identify opportunities to reduce logistics expenditure without compromising service quality.
- Explain the relationship between logistics costs, profitability, and customer satisfaction.
- Analyze a simple logistics operation to identify its major cost drivers.

Key terms

Total Logistics Cost — The combined expenditure associated with planning, transporting, handling, storing, managing, and distributing goods.

Transportation Cost — The expense incurred in moving goods between locations.

Warehousing Cost — The expense associated with operating storage facilities and handling goods within them.

Inventory Holding Cost — The cost of maintaining inventory over time, including capital, storage, insurance, and obsolescence expenses.

Cargo Handling Cost — The expense incurred in loading, unloading, lifting, moving, and securing cargo.

Direct Cost — An expense that can be specifically attributed to a particular shipment, service, or activity.

Indirect Cost — An expense shared across multiple activities that cannot be directly attributed to a single shipment without allocation.

Fixed Cost — An expense that remains relatively unchanged within a relevant operating range over a specified period.

Variable Cost — An expense that changes according to the level of activity or service provided.

Cost Driver — A factor that influences the amount of expenditure incurred by a logistics activity.

Cost Allocation — The process of assigning shared expenses to specific operations, customers, or services.

Operating Margin — The proportion of revenue remaining after deducting operating expenses, calculated according to the applicable accounting method.

Teaching content

KEY CONCEPT

Total logistics cost: a lens, not an invoice

Courses 1 to 3 treated logistics as an operational discipline: how goods are transported, warehoused, handled through ports, documented, and sold across borders under trade terms. Module 10 now attaches money to those operations, and this opening lesson supplies the map of where that money goes.

Total logistics cost is the combined expenditure associated with planning, transporting, handling, storing, managing, and distributing goods.

The first thing to understand is that total logistics cost is not a document anyone receives. No invoice arrives labelled with that title. The freight forwarder invoices freight, the terminal invoices handling, the clearing agent invoices its fee, the landlord invoices rent, the insurer invoices its premium, and the finance department records the money tied up in stock. Total logistics cost is what you see when you gather all of those separate lines into one view. It is an accounting lens: a deliberate way of looking at the business that collects every cost incurred because goods must move, wait, be handled, be cleared through borders, and be served to customers.

The lens matters because businesses make decisions with it. Prices must be set high enough to recover the full cost of getting goods to customers, not merely the freight. Profitability depends on costs that are actually understood rather than partly seen. And service promises are never free: every commitment about speed, availability, and condition is paid for somewhere in the cost structure. An organisation that watches only its freight invoices will understate what its

logistics truly costs, and it will make decisions that look excellent on one line and expensive in total.

This lesson therefore does four things. It names the cost families that make up the total. It sorts costs into direct and indirect, and into fixed and variable. It teaches total-cost thinking, the discipline of judging a decision by its effect on the whole rather than on one line. And it shows who inside a business actually owns these costs, which is rarely one person. The next lesson, Freight Costing and Landed-Cost Calculations, builds on this map to calculate landed cost properly.

KEY CONCEPT

The movement and storage families: transportation, warehousing, and cargo handling costs

The first three cost families pay for the physical work of logistics, the work every earlier module described operationally.

1. Transportation cost. Transportation cost is the expense incurred in moving goods between locations. It covers the long international legs, ocean freight with its surcharges and air freight, and the domestic distribution legs that carry goods from ports and warehouses to customers' premises. Lesson 7.4, Shipping Charges, Freight Rates, and Surcharges, took a freight invoice apart into base freight, surcharges, and local charges; under the total-cost lens, all of those lines belong to one family, and so does the fuel and maintenance of a business's own delivery fleet. Whether carriage is bought from carriers or performed in-house, the cost of movement enters the same family.

2. Warehousing cost. Warehousing cost is the expense associated with operating storage facilities and handling goods within them. It includes the space itself, whether paid as rent or borne as the cost of owning a building, and the labour of receiving, put-away, picking, packing, and dispatch that Module 8 described as the warehouse's daily work. Equipment such as forklifts and racking, utilities, security, and the systems that record stock movements all belong to this family.

3. Cargo handling cost. Cargo handling cost is the expense incurred in loading, unloading, lifting, moving, and securing cargo. It arises at every interface: cranes and terminal equipment at a port, stevedoring at a bulk berth, forklift work at a warehouse dock, lifting arrangements at a customer's premises for heavy items. Sometimes handling is invoiced separately, as when a carrier charges a terminal handling charge; sometimes it is embedded inside a wider charge or performed by the business's own staff. The lens counts it either way.

These three families are the visible backbone of logistics expenditure. They are also the lines managers argue about most, which is why the lesson later asks whether the visible backbone is where the real decisions are won.

KEY CONCEPT

Holding stock and running the paperwork: inventory holding and administrative costs

The next two families are quieter than freight and rent, and one of them never appears on an invoice at all.

1. Inventory holding cost. Inventory holding cost, also called inventory carrying cost, is the cost of maintaining inventory over time, including capital, storage, insurance, and obsolescence expenses. Its elements deserve to be named separately. Capital cost is the money tied up in the goods themselves: cash converted into stock is cash that cannot repay loans, fund other purchases, or earn returns elsewhere. Insurance is the premium that protects the stock while it sits. Obsolescence and deterioration are the losses that accumulate as goods age out of fashion, pass expiry dates, corrode, or otherwise lose value with time. Shrinkage is the gradual disappearance of stock through theft, damage, or record errors. Note the boundary with the previous family: warehousing cost pays for the facility, while inventory holding cost is largely the cost of the stock sitting inside it, and it grows with the value of the goods and the time they are held. These are different families even though they share a building.

2. Order processing and administration. Every order raised, booking confirmed, customs declaration prepared, and invoice matched consumes administrative effort: communication with suppliers, carriers, agents, and customers; data entry and record keeping; and the systems that support them. Each single transaction is inexpensive, but administration scales with the number of orders, shipments, and documents an operation generates, so an operation built on many small orders carries a proportionally heavier administrative burden than one built on few large ones.

Professionals learn to distrust the quiet families. Transportation and warehousing announce themselves on invoices; holding and administration must be sought out, which is exactly why businesses that never look for them systematically underestimate their total logistics cost.

KEY CONCEPT

Crossing borders, protecting goods, and the costs of failure

The remaining families complete the picture, and two of them concern failure rather than flow.

1. Customs and trade-compliance costs. For cross-border trade, duties, taxes, and levies are a genuine cost of trading, imposed by law and driven by how goods are classified. In Ghana, an import assessment consists of import duty plus additional taxes and levies, all computed when ICUMS, the country's single-window e-customs platform, assesses the declaration, with payment made at a participating bank. This lesson teaches the categories and their behaviour, never amounts; the closing note of this lesson returns to that rule and its reasons. Alongside the government charges sit brokerage and agency fees: the professional fee of the licensed clearing agent who lodges the declaration and arranges duty payment, and, depending on the goods, the

cost of permits and certificates from regulatory agencies such as the Food and Drugs Authority for food and medicines or the Ghana Standards Authority for standards conformity.

2. Packaging and protecting goods. Protective packaging, pallets, crating for machinery, and labelling all cost money, and so do the insurance premiums that protect cargo in transit. Packaging expenditure is not only a materials cost; it is a damage-prevention cost, and skimping on it usually shows up later in the failure families below.

3. Quality failure costs. Damaged goods, lost goods, returns, re-shipments, and the administration of claims are logistics costs caused by things going wrong. Reverse logistics, the flow of goods back toward their origin, is part of logistics rather than an exception to it, and it consumes transport, handling, and administrative effort every time it runs.

4. Stockout costs. The final family is the cost of not having stock when a customer wants it: sales lost to competitors, customers lost to disappointment, production stopped for want of materials, and expensive expediting to recover. No invoice records a stockout, but the lesson treats it seriously precisely because it is a real, sometimes severe, cost that hides completely.

A complete cost picture therefore spans flow, storage, administration, compliance, protection, and failure. Missing any family distorts every decision built on the picture.

KEY CONCEPT

Direct and indirect costs

The first great sorting of logistics costs asks a simple question about each expense: can it be attached to one specific shipment, service, or activity?

A direct cost is an expense that can be specifically attributed to a particular shipment, service, or activity. The ocean freight on a named consignment is direct: it belongs to that consignment and no other. So are the fuel consumed on one delivery run, the clearing fee charged for one declaration, and the special crating bought for one machine.

An indirect cost is an expense shared across multiple activities that cannot be directly attributed to a single shipment without allocation. The warehouse manager's salary serves every pallet in the building, not one order. The rent, the facility insurance, the telephone system, and the general software licences support the operation as a whole. No document will ever tell you which consignment caused them.

Cost allocation is the process of assigning shared expenses to specific operations, customers, or services. Because indirect costs are real, businesses that want to know what a product, a customer, or a delivery lane truly costs must spread the shared expenses across them using reasoned bases: floor space occupied, weight or volume carried, number of orders or lines handled, or hours of equipment used. Allocation is never perfect, and different reasonable bases give different answers, which is why allocation choices should be stated openly rather than buried.

The distinction matters operationally. Direct costs answer questions about individual shipments: what did this consignment cost to deliver? Indirect costs answer questions about the operation as a whole: what does our logistics infrastructure cost per period, and how should that burden be shared? A quotation built only from direct costs will underprice, because it ignores the share of the warehouse roof that every order uses. A manager who treats an indirect cost as if it were direct, and charges it wholly to one unlucky customer, will misread that customer's profitability. The next lesson's landed-cost arithmetic works with direct costs; the total-cost lens needs both kinds.

KEY CONCEPT

Fixed and variable costs

The second sorting asks a different question about each expense: does it change when the level of activity changes?

A fixed cost is an expense that remains relatively unchanged within a relevant operating range over a specified period. Warehouse rent is fixed: the same amount falls due whether the warehouse ships one hundred orders a month or a thousand, provided the operation stays within the range the facility can handle. Salaried supervisors, equipment lease instalments, and software subscriptions behave the same way. The qualifier matters: fixed costs are fixed within a relevant range and period. Outgrow the warehouse and the rent steps up; the term is relative, not absolute.

A variable cost is an expense that changes according to the level of activity or service provided. Fuel rises with kilometres driven, casual handling labour rises with tonnes moved, packaging materials rise with orders dispatched, and carrier charges rise with shipments booked. At zero activity most variable costs fall toward zero; fixed costs remain.

Many real costs are mixed. A warehouse combines fixed rent with variable handling labour; a delivery fleet combines fixed vehicle finance with variable fuel and maintenance; a clearing agent's invoice may combine a fixed per-declaration fee with disbursements that vary with the shipment. Experienced managers therefore ask not whether a cost is fixed or variable in principle, but what drives it and over what range.

The sorting earns its keep in decision-making. Fixed costs reward utilisation: once the rent is paid, every additional pallet through the same building lowers the fixed cost carried by each pallet, so capacity that sits idle is capacity quietly bleeding. Variable costs follow volume decisions: every extra order adds its own incremental cost, which is why the marginal order, not the average, should be compared against the marginal revenue. And the distinction feeds the trade-off thinking later in this lesson, because many logistics decisions are precisely exchanges of fixed for variable cost, or of one variable cost for another.

KEY CONCEPT

Cost drivers and the classic trade-offs of total-cost thinking

A cost driver is a factor that influences the amount of expenditure incurred by a logistics activity. Distance drives transport cost; time held and value drive inventory holding cost; order count drives administration; errors drive failure costs. Total-cost thinking is the discipline of identifying the drivers behind each family before optimising any of them, because the drivers are connected: pull one down and another rises.

The fallacy this discipline dismantles is optimising a single line. The cheapest transport offer can mean slower transit, which means more stock in transit and on the shelf, more capital tied up, and more expediting when demand outruns the slower pipeline. The cheapest warehouse can mean careless picking, which means more returns, more claims, and more re-shipping. In both cases the saving is real but the cost it triggers elsewhere is larger, and the business that compares only the invoice lines will congratulate itself into losses.

Three classic trade-offs recur throughout logistics.

1. Transport cost versus inventory cost. Faster, dearer transport shortens the pipeline and reduces the stock that must be held to cover the time goods spend moving and waiting; slower, cheaper transport does the reverse. Neither mode is right in itself; the choice depends on the value of the goods, the demand pattern, and what stockouts would cost.

2. Warehouse count versus delivery cost. More, smaller warehouses place stock closer to customers and shorten the final delivery legs, improving service and cutting outbound transport; but they multiply facilities, duplicate stock in several places, and raise total holding cost. Fewer, larger warehouses do the opposite. Network design is this trade-off made concrete.

3. Order size versus carrying cost. Ordering in large lots keeps ordering and administrative effort low but holds average stock high, raising holding cost; ordering in small lots does the reverse. The classic economic order quantity model balances the two pressures to find the order size that minimises their combined total, and its logic, in words, is exactly this balancing act. Treat it as a model, not a law: it assumes steady demand and simple costs, and Lesson 8.3, Inventory Control, FIFO, and FEFO, has already supplied the reorder-point and safety-stock ideas with which it works in practice.

The lesson of all three trade-offs is the same: the question is never which line can be made smallest, but which combination of lines can be made smallest together without harming service.

KEY CONCEPT

Visible and hidden costs: the invoice and the tail

Sort the same cost families by how they announce themselves and a second discipline emerges.

Visible costs are the ones that arrive on documents: the freight invoice, the warehouse rent, the clearing fee, the insurance premium. They are easy to see, easy to budget, and therefore the lines that attract nearly all managerial attention.

Hidden costs are the ones embedded in operations without a dedicated document. Dwell time, the hours and days cargo waits in yards and terminals, consumes space and equipment and the importer's patience while appearing on no invoice until it overflows into charges. Demurrage, detention, and terminal storage, introduced in Module 9's treatment of port storage and to be calculated in this module's Lesson 10.3, Storage, Detention, Demurrage, and Other Charges, are dwell time that has finally become invoiced, usually after the expensive part of the waiting has already happened. Re-work and re-handling consume labour twice for one order. Expediting turns a planning failure into premium transport. Claims administration pays office salaries to process what damage caused. None of these appears in the accounts under its own name; they scatter across wages, fuel, and write-offs.

This is why landed-cost thinking must include the tail. Landed cost is the full cost of placing goods at their final destination: purchase price, international freight, insurance, duties, taxes and levies, local charges, and inland transport, together with the tail of delay and failure costs that the journey realistically generates. A landed cost built only from quoted lines is a forecast of the best case, not a measurement of the operation. Professionals add the realistic tail: the probable waiting, the occasional expediting, the share of shipments that will be damaged and re-served.

The next lesson formalises the landed-cost calculation line by line. The work of this lesson is the habit that makes the calculation honest: after listing what will certainly be invoiced, ask what else this decision makes likely, and price that likelihood too.

KEY CONCEPT

Who owns logistics cost

If total logistics cost is a lens, who is supposed to look through it? In most businesses, no one person does, and that is precisely the management problem.

Procurement controls how goods are bought: the agreed Incoterms 2020 rule in the sales contract decides who arranges and pays for carriage, insurance, and clearance, so the buyer's logistics cost is partly fixed before its logistics department is ever consulted. The warehouse team owns space, labour, and equipment spending. The transport team owns fleet, fuel, and carrier purchases. Finance owns the money tied up in inventory, the insurance arrangements, and the payment of duties and levies, and it sees the aggregate bills that nobody else views whole. Sales, meanwhile, makes service promises whose costs land in all four of the other functions.

Each function therefore sees one slice of the total, and each can optimise its slice at the others' expense. Procurement saves money by buying in huge lots; the warehouse drowns in stock and

the holding cost rises. Transport chooses the cheapest, slowest carrier; expediting and stock costs rise somewhere no transport manager is measured on. A business that evaluates each function only on the costs it directly controls will systematically reward decisions that raise the total.

Assembling a true total-cost view therefore requires coordination: shared figures, agreed allocations, and decisions taken across functions rather than within them. Cost allocation, introduced earlier in this lesson, is the practical tool, because shared expenses must be spread before anyone can see what a customer, product, or lane truly costs. The full strategic treatment of coordinating logistics decisions, and of choosing between doing the work in-house and buying it from service providers, belongs to Lesson 15.1, Logistics Strategy and Service-Provider Selection; here, the point is the diagnosis. When a total refuses to fall although every line manager insists their line has been cut, look for the boundary between functions: the saving is usually being paid for in the department next door.

KEY CONCEPT

Reducing cost without damaging service: profitability and customer satisfaction

Reducing logistics expenditure without compromising service quality is a legitimate goal, and the cost families point to where it is honestly found.

1. Raise utilisation. Fill vehicles and containers properly, consolidate small loads, and keep warehouse capacity working. Higher utilisation spreads fixed costs across more output, which lowers unit cost without touching service.

2. Cut avoidable waiting. Fast, accurate documentation, early declaration lodgement, and disciplined coordination with terminals keep cargo inside its free time and out of the delay charges. Waiting removed is cost removed, and service usually improves in the same motion.

3. Right-size inventory. Better forecasting, disciplined reorder points, and order sizes chosen with holding cost in view reduce the capital, insurance, and obsolescence burden. Lesson 8.3's stock-control tools are the operating instruments of this lever.

4. Reduce errors. Accurate picking, careful handling, and sound packaging cut the returns, claims, and re-shipments of the failure families. Quality spending is often the cheapest cost reduction available, because every error prevented removes several downstream costs at once.

5. Buy intelligently. Blending contracted rates for predictable volumes with spot purchases for peaks, and comparing offers on total cost rather than headline rate, disciplines the largest single family without a single operational change.

What all five levers share is that they lower cost while preserving or improving what the customer experiences. A cut that slows deliveries, thins availability, or increases damage is not a saving; it is a transfer of cost to the customer, and it returns as lost revenue.

That return path is the connection to profitability. Operating margin is the proportion of revenue remaining after deducting operating expenses, calculated according to the applicable accounting method, and logistics cost sits directly inside that deduction: every cedi or dollar removed from avoidable logistics cost flows toward the margin. Customer satisfaction works on the other side of the account. Reliable, accurate, available service retains customers and generates repeat revenue; service failures quietly drain it. Logistics cost and customer service are therefore not opponents to be balanced once, but a pair to be managed continuously: the purpose of the total-cost lens is to find the money that is being spent without buying either efficiency or service.

WORKED EXAMPLE

One machinery consignment, many cost lines: a Kumasi importer's total-cost picture

A machinery dealer in Kumasi imports one consignment of industrial equipment, shipped by sea and discharged at Tema, one of Ghana's two seaports, which are managed by the Ghana Ports and Harbours Authority. All figures in this example are illustrative for teaching purposes; they show how a total-cost picture is assembled, not what any real shipment costs.

Step 1: Fix the frame. The agreed Incoterms 2020 rule in the sales contract determines which of the coming lines fall to the importer's account. Here the rule leaves carriage, insurance, and import clearance to the buyer, so every line below belongs to the Kumasi dealer's total-cost picture.

Step 2: The movement lines. Ocean freight on the equipment, including fuel-related and other surcharges, is invoiced at an illustrative 2,400 US dollars. Cargo insurance for the transit leg costs an illustrative 150 US dollars.

Step 3: The compliance lines. The dealer's licensed clearing agent lodges the declaration in ICUMS. The system assesses import duty plus additional taxes and levies, which for this illustration total 18,000 GHS, paid at a participating bank. The agent's professional fee is 3,000 GHS. These amounts are purely illustrative: real amounts depend on tariff classification and the law in force, and this course never teaches figures for them.

Step 4: The local lines. Terminal handling, documentation, and other local charges at the port come to an illustrative 4,500 GHS, and road carriage from Tema to Kumasi adds an illustrative 6,500 GHS.

Step 5: The tail, deliberately zero. Demurrage, detention, and terminal storage for this consignment are zero, and the zero is a result, not luck. The agent lodged documents early, the assessment was paid promptly, and the container left the terminal within the free time allowed by the carrier and terminal. On a consignment where clearance had drifted past that point, the same lines could have carried substantial charges; here, fast clearance kept the entire tail at zero.

Step 6: The invisible lines. No invoice names them, but the picture is not complete without them: the money tied up in the duty payment and in equipment awaiting sale, insurance on the goods in store, and the dealer's staff time across the transaction.

What does this example demonstrate?

The freight invoice, 2,400 US dollars, was the number the dealer's accountant would once have called the logistics cost. Assembled through the lens, the picture holds five families, two currencies, one deliberately zero tail, and a set of invisible lines. Key lesson: a total-cost picture for one import is a ledger assembled from many documents and a few honest estimates, and the assembler who stops at the freight invoice has read perhaps a fifth of the book.

WORKED EXAMPLE

Air versus sea for spare parts: reading the trade-off honestly

A service company in Singapore maintains industrial equipment for customers across the region and sources spare parts from a supplier in Hamburg. Management must choose between replenishing them by sea and by air. All figures in this example are illustrative for teaching purposes.

Step 1: Price the two pipelines. By sea, freight plus surcharges and local charges at both ends cost an illustrative 400 US dollars per shipment, with a door-to-door lead time of several weeks. By air, the equivalent charge is an illustrative 1,800 US dollars per shipment, arriving within days. On the freight invoice alone, air is more than four times dearer, and the meeting proposes sea.

Step 2: Trace what lead time does to stock. Lesson 8.3 taught that a reorder point must cover demand expected during the lead time, plus safety stock against variability. A several-week pipeline therefore forces the company to hold substantially more stock in Singapore than a days-long one: more parts on the shelf while the next shipment sails.

Step 3: Price the extra stock. Holding that larger buffer costs money every cycle: capital tied up in parts, insurance, and, for parts that age or are superseded, obsolescence. For an illustration, suppose the sea pipeline's extra buffer adds 2,000 US dollars of holding cost per replenishment cycle across the part range, against 300 US dollars for the air pipeline. The sea option's apparent 1,400 US dollars of freight saving per cycle now competes against 1,700 US dollars of additional holding cost per cycle, and the arithmetic has changed sides.

Step 4: Price the stockout exposure. When a critical part fails at a customer's site, the company expedites regardless of policy, paying premium freight and risking the relationship. A short pipeline shrinks that exposure; a long one enlarges it, and the value of the difference depends on how often failures occur and what each costs, a question about the company's own history, not freight tariffs.

Step 5: Decide per part, not per policy. The honest conclusion is mixed. For heavy, low-value, predictably consumed parts, the sea pipeline's holding penalty is small and sea wins. For high-value critical parts, the air premium is small against the holding cost and stockout exposure it avoids, and air wins. A split policy, sea for the routine bulk and air for critical or high-value items, is often the true minimum.

What does this example demonstrate?

The freight comparison, four times dearer by air, was not the deciding comparison. Key lesson: transport decisions are inventory decisions in disguise; compare the total cost of each pipeline, including the stock it forces you to hold and the stockouts it permits, and let different parts ride different pipelines.

WORKED EXAMPLE

The cheap warehouse and the error tail

A distributor of fast-moving consumer goods in Accra outgrows its warehouse and compares two facilities. Warehouse A quotes a rent of 19,000 GHS per month; warehouse B quotes 23,000 GHS per month but operates with tighter picking discipline and better scanning. Management chooses A for its 4,000 GHS monthly saving. All figures in this example are illustrative for teaching purposes.

Step 1: Establish the error rates. In this illustration, warehouse A mis-picks three order lines in every hundred; warehouse B mis-picks half a line in every hundred. The distributor processes 2,000 order lines a month. Warehouse A therefore generates about 60 errors monthly, warehouse B about ten: fifty extra errors every month, purely illustrative rates for a purely illustrative pair of facilities.

Step 2: Price one error's tail. Each picking error triggers a small chain: a re-delivery or return journey, say 300 GHS of transport; re-picking, re-packing, and claims handling, say 200 GHS of labour and administration; and, when goods are wrong or late, a credit note or replacement that erodes the month's margin. At an illustrative 500 GHS of direct tail per error, fifty extra errors cost about 25,000 GHS per month.

Step 3: Compare the totals. Warehouse A's rent saving was 4,000 GHS per month. Its error tail costs roughly 25,000 GHS per month. The cheap warehouse is the expensive one by around 21,000 GHS per month, before counting the quiet losses: customers who quietly reduce their orders after repeated wrong deliveries, and staff time spent apologising rather than selling. Those losses are real but never appear under the heading "warehouse error cost", which is why the cheap option survived the original comparison.

Step 4: Ask why the comparison failed. The rent was visible, invoiced, and easy to compare; the error tail was hidden, scattered across transport, wages, and lost revenue, and attached to no single budget. The distributor compared the only two numbers it could see and mistook them for the whole decision.

What does this example demonstrate?

Quality failure costs are the classic hidden family: small per error, frequent in aggregate, and invisible in the accounts. Key lesson: before signing for the cheaper facility, price the tail, the returns, claims, re-ships, and eroded trust that its error rate makes likely, and compare the rent saving against the whole of what the saving buys.

IMPORTANT NOTE

The cheapest invoice is not the cheapest logistics

The misconception this lesson has been dismantling throughout deserves one final, blunt statement: the cheapest invoice is not the cheapest logistics. A decision that lowers one cost line can raise the total, because the lines are connected through the cost drivers. Cheaper transport usually means slower transport, and slower transport means more inventory holding cost and more stockout exposure. A cheaper warehouse usually means a less careful warehouse, and carelessness means returns, claims, and re-shipping. A larger order size means fewer ordering efforts but more money sleeping on shelves.

The corrective is not a rule of thumb but a habit of comparison: when an offer lowers a visible line, ask which other lines this offer changes, find where they are recorded or estimate them honestly, and compare the offer's full effect on the total against the alternative's full effect. Sometimes the cheap option survives that comparison intact, because the hidden lines genuinely do not move. The discipline is not that cheap is wrong; it is that cheap must be proven against the total, not declared against the line. A saving that costs more elsewhere than it saves where you are looking is not a saving at all, whatever the invoice says.

IMPORTANT NOTE

Inventory carrying cost is real money even though no invoice names it

Learners trained on invoices tend to believe that a cost exists only when a document charges it. Inventory holding cost is the standing refutation. When stock sits in a warehouse, no invoice arrives for holding it, yet real expenditure accumulates continuously: the capital tied up in the goods is money not repaying loans or funding the business; insurance premiums protect it; obsolescence and deterioration quietly reduce its value; shrinkage removes pieces of it. All of this is genuine cost, and on high-value stock held for a long time it can rival the freight that delivered the goods.

The reason the cost escapes notice is bookkeeping, not economics. Its elements are scattered: financing cost inside finance charges, insurance inside insurance premiums, deterioration inside write-offs, shrinkage inside stock discrepancies. Nothing in the accounts aggregates them under the name "holding cost", so a manager reading the accounts sees only fragments. The professional response is to assemble the fragments deliberately, as this lesson's worked examples do, and to treat the assembled figure as a first-class cost in every decision about order

sizes, stock levels, and transport speed. Money does not need an invoice to be spent; it needs only to be committed, and holding stock commits it.

IMPORTANT NOTE

Duty and levy amounts are a matter of classification and current law: this lesson teaches categories, never figures

Throughout this lesson, customs charges have been named as categories and never as amounts, and the restraint is absolute, not stylistic. Duty, tax, and levy amounts depend on how goods are classified in the tariff, on the value customs accepts, and on the law in force at the time of assessment, all of which change. In Ghana, the assessment is computed when ICUMS processes the declaration, and its composition follows the tariff treatment of the specific goods on the specific day. Any figure memorised from a lesson would be wrong somewhere, soon, and expensively.

The professional habits this lesson can teach are the ones that do not expire: know which family a charge belongs to; know that the licensed clearing agent lodges the declaration and the assessment is paid against the bill the system generates; know that classification and valuation drive the amounts, which is why both are treated as serious technical work elsewhere in this course; and know that the landed-cost method, built in the next lesson, always takes its duty and levy figures from the live assessment, never from memory. When you need an amount, you consult the current tariff position for the goods or the assessment itself. When you teach or explain, you name the categories and point to the source. That is the discipline of a cost professional, and it begins here.

Practice questions (Lesson Checks)

Practice Question 1

A purchasing manager at a manufacturing company in Takoradi selects the lowest ocean freight quote for its monthly raw-material imports. The cheaper service has a noticeably longer transit time. Three months later, the finance officer reports that overall logistics spending has risen, driven by higher warehouse stock levels and occasional urgent air shipments arranged to cover production shortages.

Which option best explains why total logistics cost rose even though the freight line was reduced?

1. The cheaper, slower service lengthened the lead time, so the company had to hold more inventory to cover the longer pipeline and pay for expediting when demand outran stock; those increases outweighed the freight saving. ✓

2. The finance officer has double-counted, because total logistics cost is the sum of the freight invoices and the freight invoices fell.

3. Warehouse stock levels rose because the cheaper carrier must have damaged more cargo, and damaged cargo always increases inventory.

4. Ocean freight quotes are inherently unreliable, so choosing the lowest offer is always a mistake regardless of transit time or inventory effects.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

This is the classic transport-versus-inventory trade-off taught through total-cost thinking. The cheaper service's longer transit lengthened the lead time, and a longer lead time raises both the stock that must be held to cover demand during the pipeline and the exposure to stockouts, which the company experienced as urgent air shipments. The freight saving was real, but the cost it triggered in the inventory and failure families was larger, so the total rose.

Answer 2 fails because it restates the very misconception the lesson dismantles: total logistics cost is not the sum of freight invoices; it spans holding, administration, compliance, failure, and stockout costs as well. Answer 3 invents a mechanism; the report attributes the increase to higher stock levels and expediting, not to damage, and slower transit affects inventory through lead time, not through condition. Answer 4 over-generalises in the opposite direction: the cheapest offer is wrong only when its hidden effects outweigh the saving, which the discipline of comparing totals reveals case by case.

The takeaway: judge any cost reduction by its effect on the whole lens, never by the line it improves.

Practice Question 2

A logistics manager in Accra is asked to classify four expenses: the monthly salary of the warehouse supervisor, the fuel consumed on each delivery run, the ocean freight charged for one specific consignment, and the annual rent of the warehouse building.

Which option best describes how these costs should be classified?

1. All four are variable costs, because every cost changes when the business grows.

2. All four are direct costs, because all of them appear on invoices or payroll records received by the company.

3. The fuel per delivery run and the freight for the specific consignment are direct and variable, while the supervisor's salary and the warehouse rent are indirect and largely fixed within the company's current operating range. ✓

4. The salary and rent are direct costs of the warehouse, while the fuel and freight are indirect costs shared across all shipments.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The two sortings answer different questions. Direct versus indirect asks whether the expense can be attributed to one specific shipment, service, or activity: fuel burned on one delivery run and freight invoiced for one named consignment can, while the supervisor's salary and the building's rent support the whole operation and can only be shared through cost allocation. Fixed versus variable asks whether the expense changes with activity: fuel and shipment charges rise and fall with volume, whereas salary and rent remain relatively unchanged within a relevant operating range over the period.

Answer 1 fails because salary and rent are the textbook fixed costs; growth eventually steps them up, but within the current range they do not vary. Answer 2 confuses visibility with directness: appearing on an invoice does not make a cost attributable to one shipment, which is exactly why allocation exists. Answer 4 reverses both labels, treating the shared, range-stable expenses as attributable and the attributable, activity-driven ones as shared.

The takeaway: ask two separate questions of every cost, who or what it belongs to, and whether it moves with activity; the two answers together place any expense correctly.

Practice Question 3

A distributor in Kumasi keeps a large stock of finished goods in its own warehouse. The managing director argues that the stock costs the company nothing while it sits, because the supplier has already been paid, the warehouse is already built, and no invoice arrives for goods the company owns.

Which statement best explains the flaw in this argument?

1. Holding costs apply only to rented warehouse space; goods a company owns and stores in its own building incur no further cost.

2. Inventory holding cost is real expenditure even without a dedicated invoice: capital tied up in the stock, insurance, obsolescence and deterioration, and shrinkage all accumulate while goods are held, even though the accounts record them only as scattered fragments. ✓

3. The stock costs nothing provided it is sold before the end of the financial year, because holding costs are only charged after twelve months.

4. Inventory holding cost is a government charge collected through customs on goods held in warehouses after importation.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Inventory holding cost is the cost of maintaining inventory over time, including capital, storage, insurance, and obsolescence expenses. None of these requires an invoice addressed to the stock. The cash converted into goods is money unavailable for other uses; the insurance premium protects the goods; value drains away as goods age, expire, or fall out of demand; and shrinkage removes part of the quantity. The accounts scatter these fragments across finance charges, insurance premiums, write-offs, and stock discrepancies, which is precisely why a manager reading only invoices concludes the stock is free.

Answer 1 fails because ownership of the building removes the rent line only, while the capital, insurance, obsolescence, and shrinkage elements continue regardless of who owns the shelves. Answer 3 invents a rule that does not exist; holding cost accrues from the first day, not after any anniversary. Answer 4 confuses a business's internal cost of holding stock with government charges at importation, which belong to the customs family and follow the assessment, not the storage.

The takeaway: no invoice is needed for money to be spent. Committed capital and accumulating risk are costs, and the total-cost lens exists to make them visible.

Fact-check appendix (audit notes — not for entry)

Definitions of total logistics cost, transportation cost, warehousing cost, inventory holding cost, cargo handling cost, direct cost, indirect cost, fixed cost, variable cost, cost driver, cost allocation, and operating margin follow the lesson brief's key terms, defined at first use and used in the body. — source: lesson brief 10.1, Key Terms (curriculum database).

Total-cost trade-off principle: slower, cheaper transport raises in-transit and cycle inventory and stockout exposure; faster transport costs more per unit but cuts inventory and stockouts. — source: research sheet logistics-scm-frameworks.md, section 6.

EOQ is the classic textbook model balancing ordering and holding cost, taught in words with its strong simplifying assumptions (steady demand, fixed costs), as a model not a law; no formula stated per tasking. — source: research sheet logistics-scm-frameworks.md, section 5 and DO NOT CLAIM item 18.

Reorder point covers expected demand during lead time plus safety stock; safety stock buffers demand and lead-time variability. — source: research sheet logistics-scm-frameworks.md, section 5.

Network design trade-off: more/deeper warehouse networks cut delivery time and outbound transport but add facility and inventory cost; centralisation does the reverse. — source: research sheet logistics-scm-frameworks.md, section 8.

Reverse logistics (returns flows back toward origin) is part of logistics management. — source: research sheet logistics-scm-frameworks.md, section 1.

Capacity utilisation as an efficiency lever for cost; consolidation of loads; blending contract rates for committed volume with spot purchases for peaks; freight cost per unit as the standard comparator. — source: research sheet logistics-scm-frameworks.md, sections 6, 7 and 10.

Dwell-time-into-charges concepts: free time is the allowance before time-based charges begin; demurrage (carrier, container inside terminal beyond free time), detention (carrier, container outside terminal not returned), storage (terminal's own space charge); rates and free days are purely contractual tariffs, so this lesson names concepts only and defers calculations to Lesson 10.3. — source: research sheet trade-transport-standards.md, section 11.

Freight quote composition concept: base ocean freight plus surcharges and charges such as terminal handling at both ends, documentation, and security charges; names only, no amounts. — source: research sheet trade-transport-standards.md, section 7.

Cargo insurance as a purchased protection with premium, taught generically; no cover levels or percentages. — source: research sheet trade-transport-standards.md, section 10.

Incoterms 2020 allocate tasks, costs, and risk-transfer points between buyer and seller and decide who arranges and pays for carriage, insurance, and clearance; they do not fix price or transfer ownership. Used qualitatively in the Kumasi example's Step 1. — source: research sheet trade-transport-standards.md, section 1.

In Ghana, an import assessment consists of import duty plus additional taxes and levies, all computed when ICUMS assesses the declaration, paid at participating banks; no rates, percentages, or named levy components stated anywhere. — source: research sheet ghana-trade-customs.md, sections 1 and 7 (safe phrasing).

Importers work through licensed customs clearing agents, who lodge the declaration in ICUMS and arrange duty payment. — source: research sheet ghana-trade-customs.md, section 6 (safe phrasing).

Depending on the goods, importers may need permits from regulatory agencies such as the Food and Drugs Authority (FDA) for food and medicines or the Ghana Standards Authority (GSA) for standards conformity. — source: research sheet ghana-trade-customs.md, section 5 (safe phrasing).

Ghana's two seaports, Tema and Takoradi, are managed by the Ghana Ports and Harbours Authority (GPHA). — source: research sheet ghana-trade-customs.md, section 4 (safe phrasing).

ICUMS is Ghana's single-window e-customs platform (description only). — source: research sheet ghana-trade-customs.md, section 1 (safe phrasing).

Stockout consequences (lost sales, expediting, production stoppages, eroded customer confidence) taught qualitatively with no magnitudes. — source: general business knowledge.

Picking-error chains (re-delivery, re-pick, claims, credit notes, eroded trust) and the scattering of holding-cost elements across finance, insurance, write-offs, and discrepancies are general business knowledge taught qualitatively. — source: general business knowledge.

All companies, quantities, error rates, and monetary figures in the three worked examples (the Kumasi machinery consignment, the Singapore-Hamburg spare-parts comparison, and the Accra warehouse comparison) are illustrative for teaching purposes and are labelled as such in each example; no real rate, duty amount, levy amount, rent level, freight level, or benchmark percentage is claimed. — source: WRITING_SPEC.md fact-safety rule 5.1; the research sheets' DO NOT CLAIM lists.

References to Lesson 7.4 (freight charges anatomy), Lesson 8.3 (reorder points and safety stock), Module 8 warehousing operations, Module 9 port storage concepts, Lesson 10.2 (landed-cost calculation), Lesson 10.3 (demurrage, detention, storage calculations), and Lesson 15.1 (Logistics Strategy and Service-Provider Selection) are by name only, with no re-teaching. — source: WRITING_SPEC.md section 7; lesson briefs 7.4, 8.3, 10.2, 10.3, and module 15 titles.

Lesson 10.2 — Freight Costing and Landed-Cost Calculations

Summary

This lesson introduces the methods used to calculate transportation expenses and determine the total landed cost of imported goods. Learners will examine freight pricing methods, chargeable weight, cargo volume, insurance, duties, taxes, and destination charges. The lesson also explains how Incoterms and commercial arrangements influence which expenses should be included in a particular party's landed-cost calculation.

Learning objectives

- Define freight costing and landed cost.
- Identify the expenses commonly included in landed-cost calculations.
- Explain how freight charges are calculated using weight, volume, or container-based rates.
- Calculate chargeable weight using specified freight conversion factors.
- Distinguish between customs value and total landed cost.
- Explain how Incoterms influence responsibility for transportation and related expenses.
- Calculate the landed cost of a shipment using supplied cost and tax information.
- Identify how exchange-rate fluctuations affect import costs.
- Explain how accurate costing supports pricing and purchasing decisions.

Key terms

Freight Costing — The process of determining the expenses associated with transporting goods.

Landed Cost — The total cost of acquiring and bringing goods to a specified destination, including the expenses defined by the applicable costing policy.

Customs Value — The value determined for customs purposes under the applicable customs valuation rules.

Cost, Insurance and Freight (CIF) — An Incoterms rule under which the seller pays specified costs, freight, and insurance to the named destination port while risk transfers earlier at shipment.

Free on Board (FOB) — An Incoterms rule under which the seller delivers goods on board the vessel at the named port of shipment.

Chargeable Weight — The weight used to calculate freight charges under the applicable transport pricing rules.

Volumetric Weight — A calculated weight based on shipment dimensions and an applicable conversion factor.

Freight Rate — The price charged for transporting goods according to an agreed pricing basis.

Import Duty — A customs duty imposed on imported goods under applicable tariff and customs laws.

Import Tax — A tax imposed on imported goods under applicable national legislation.

Exchange Rate — The rate at which one currency is exchanged for another.

Cost Breakdown — An itemised presentation of the expenses contributing to a total cost.

Unit Landed Cost — The landed cost allocated to one unit of an imported product.

Teaching content

KEY CONCEPT

What landed cost means

Lesson 10.1, Components of Total Logistics Cost, mapped the families of expense that a logistics operation generates: transportation, warehousing, inventory holding, cargo handling, administration, insurance, and distribution. This lesson points that cost-family lens at a single consignment and asks the question every importer must answer: what did it really cost to get these goods from the seller's door to my warehouse, ready for sale or use?

The answer is called the landed cost. The landed cost is the total cost of acquiring and bringing goods to a specified destination, including the expenses defined by the applicable costing policy. In plain terms, it is the price of the goods plus every charge needed to move them, insure them, clear them through the border, handle them at the destination, and carry them to the buyer's premises, together with the quiet costs at the end of the journey.

The usual families in a landed-cost build for an import are these: the goods value on the supplier's invoice; the international freight; cargo insurance; duties, taxes, and levies; terminal and handling charges at the destination port or airport; inland transport to the warehouse; and a cost tail of clearance fees, storage exposure, and contingencies. Lesson 10.1 taught the families; this lesson prices one shipment through all of them, line by line.

The phrase "defined by the applicable costing policy" matters. Which lines belong in a particular party's build is not a matter of taste: it follows from the commercial arrangement between buyer and seller, above all the Incoterms rule in the sales contract, treated later in this lesson. A landed cost is always somebody's landed cost, counted from that party's door.

Landed cost matters for three decisions above all: setting a selling price that truly covers what the goods cost, choosing between suppliers in different source markets, and telling

management the honest truth about margins. A buyer who knows only the invoice price knows only the first slice of the number.

KEY CONCEPT

Freight costing and the freight rate

Freight costing is the process of determining the expenses associated with transporting goods. Its central instrument is the freight rate: the price charged for transporting goods according to an agreed pricing basis. A freight rate is never just a number; it is a number attached to a basis, and the basis decides how the arithmetic runs. "GHS 600" means nothing on its own; "GHS 600 per tonne or per cubic metre, whichever is greater" is a complete pricing rule.

The common pricing bases are these.

1. By weight. Dense, heavy cargo such as ingots or machinery castings is charged per tonne, because weight, not space, is what the cargo consumes.

2. By measurement, that is by volume. Light, bulky cargo such as empty plastic ware or expanded foam is charged per cubic metre, because space, not weight, is what it consumes.

3. By weight or measurement, whichever is greater. Ocean carriers carrying less-than-container-load (LCL) cargo combine the two bases into one rule, examined in the next block.

4. By container. A shipper with a full container load (FCL), met in Lesson 7.3, Containerisation: FCL, LCL, and Container Types, usually pays a rate for the box, not for the contents: a rate per twenty-foot or forty-foot container whether the box travels full, heavy, or light.

5. Per kilogram of chargeable weight. Air cargo prices by the kilogram, but the kilogram that counts is a calculated one, covered in its own block below.

Lesson 7.4, Shipping Charges, Freight Rates, and Surcharges, explained how a freight quotation is assembled: a base rate plus surcharges and ancillary charges such as terminal handling, documentation, and security fees. This lesson does not repeat that assembly. Here the freight quotation arrives as one number for the consignment, and the task is to apply the pricing basis behind it and fold the result into a landed cost.

KEY CONCEPT

Weight or measurement: the rule for LCL ocean freight

When cargo does not fill a container, the carrier or consolidating forwarder collects many shippers' consignments into one box. Each consignment then pays its share of the container, and the share is settled by the weight-or-measurement rule, in the trade simply called W/M: the consignment is charged on its weight in tonnes or on its volume in cubic metres, whichever gives the carrier the greater revenue. The chargeable unit that results is often called the revenue tonne: one tonne of weight or one cubic metre of space, whichever is being paid for.

The logic is fairness of a hard kind. A vessel earns revenue from space and is limited by payload. A consignment of machinery castings that weighs 8 tonnes but occupies 2 cubic metres consumes the ship's weight allowance, so it pays on 8. A consignment of ping-pong balls that weighs 100 kilograms but occupies 5 cubic metres consumes the ship's space, so it pays on 5. As an illustrative shape of the rule: a consignment measuring 3 cubic metres and weighing 2 tonnes is charged on 3, because 3 is the greater; the same goods turned around, measuring 2 cubic metres and weighing 3 tonnes, would be charged on 3 again, this time by weight.

Two practical consequences follow. First, a shipper can often lower the chargeable quantity by better packing: smaller cartons and tighter stacking reduce the measurement on which an LCL consignment is charged. Second, once a shipper's cargo grows to fill a container, the arithmetic changes shape entirely: above a break-even fill, the flat rate for the whole box costs less than paying by revenue tonne, which is why consolidators quote both and shippers compare. That comparison belongs to freight negotiation; this lesson takes the quoted rate and runs the numbers it produces.

KEY CONCEPT

Chargeable weight and volumetric weight in air freight

An aircraft is scarce in two dimensions at once: it can carry only so many kilograms of payload and only so many cubic metres of volume. A consignment of tungsten drill bits would exhaust the weight allowance long before filling the hold; a consignment of feather pillows would fill the hold while barely registering on the scales. Air cargo resolves the tension with a calculated weight.

Volumetric weight is a calculated weight based on shipment dimensions and an applicable conversion factor. Under the IATA standard factor, the volumetric weight in kilograms is the length multiplied by the width multiplied by the height of the piece, all in centimetres, divided by 6,000. In other words, the standard treats every 6,000 cubic centimetres as one chargeable kilogram, which is the same as treating one cubic metre as about 166.7 kilograms. Some contracts or carriers publish different divisors, but 6,000 is the IATA standard and the factor this course uses.

Chargeable weight is the weight used to calculate freight charges under the applicable transport pricing rules. In air freight it is the higher of the actual gross weight and the volumetric weight. The airline weighs the cargo, measures it, computes both figures, and charges on whichever is greater: the drill bits pay on their real weight, the pillows pay on the air they displace.

The freight charge then follows mechanically: chargeable weight multiplied by the per-kilogram rate quoted for the lane, plus the air-side surcharges of the kind Lesson 7.4 described. One caution belongs to the costing office: the conversion factor is a rule of the pricing agreement, so the factor specified in the applicable tariff or contract governs the calculation, and it should be checked before any figure is trusted. A worked computation follows in the third worked example of this lesson.

KEY CONCEPT

From consignment freight to freight per unit

A freight quotation covers a consignment, but a business sells cartons, bags, or pieces. To land the number where decisions are made, the consignment's freight must be divided among the units it contains: freight per unit equals the total freight charges for the consignment divided by the number of sale units carried. This freight cost per unit is the standard comparator in performance measurement, and it is the freight slice of the unit landed cost developed at the end of this lesson.

The division sounds trivial, but three habits make it honest.

1. Use the consignment's full freight, not the base rate alone. The surcharges and ancillary charges attached to the shipment, from terminal handling to documentation, all travelled with the cartons and all belong in the numerator.

2. Allocate on a sensible basis. Where the units are identical cartons, an equal split is correct. Where they differ in size or weight, allocate in proportion to what drove the charge: a consignment charged by measurement should share its freight out by volume, and one charged by weight should share by weight.

3. Watch what the result teaches. Freight per unit falls when consignments are consolidated into fuller containers, falls when packing is tightened, and rises when a half-empty box sails. Two importers buying identical goods at identical freight rates can carry very different freight per unit purely because of how they pack and consolidate.

In the first worked example below, a consignment's LCL freight of GHS 10,800 across 500 cartons puts GHS 21.60 of ocean freight inside every carton before a single other cost is counted. That per-carton figure is the freight line of the per-carton landed cost, and it is the number a pricing manager actually uses.

KEY CONCEPT

The landed-cost build and its cost breakdown

A landed cost is produced by a build: an ordered list of cost lines added from the goods value to a total. A cost breakdown is an itemised presentation of the expenses contributing to a total, and the build is simply a disciplined cost breakdown in a fixed sequence, so that nothing that will actually be paid can silently vanish. The standard lines of an import build run as follows.

1. Goods value. The price of the goods on the supplier's commercial invoice, converted into the buyer's own currency at the rate that applies when payment is made. This is the only line most quotations ever show.

2. International freight. The ocean or air charge for the consignment, computed on the pricing basis of the previous blocks.

3. Cargo insurance. The premium for covering the transit risk, whether bought by the buyer or, under some seller-paid arrangements, arriving embedded in the goods price.

4. Duties, taxes, and levies. Import duty is a customs duty imposed on imported goods under applicable tariff and customs laws; import tax is a tax imposed on imported goods under applicable national legislation, value added tax being the common example; levies are further statutory charges attached to importing. Each is computed by rule on its own base, as the next block explains. In Ghana, an import assessment consists of import duty plus additional taxes and levies, and the component names include import duty, VAT, the National Health Insurance Levy (NHIL), the GETFund levy, the ECOWAS levy, and the AU levy; the assessment is generated as a tax bill when the declaration is processed in ICUMS and is paid at a participating bank.

5. Terminal and handling charges. Destination-side charges such as the terminal handling charge, documentation, and security fees at the port or airport.

6. Inland transport. The road or rail leg from the port or airport to the buyer's warehouse.

7. Clearance and agency fees. What the licensed clearing agent and any other service providers charge for lodging the declaration and arranging the entry.

8. The cost tail. Storage, time-based charge exposure, financing, and a contingency margin, the quiet lines that an honest build refuses to ignore.

Builds differ: a policy may merge lines or split them further. What no honest build does is omit a line the buyer will really pay.

KEY CONCEPT

Customs value versus landed cost

The most consequential base in the build is the customs value: the value determined for customs purposes under the applicable customs valuation rules. It is the figure on which import duty is computed, and it is not the same animal as the landed cost.

Under the international valuation rules administered by the World Customs Organization, valuation proceeds through six methods applied in strict order, and the first method is transaction value: the price actually paid or payable for the goods when sold for export, adjusted for prescribed additions where the rules require them. Only where that method cannot be used do the fallback methods apply in sequence. The general principle for this lesson is simple: duty is computed on a customs value determined by rule, at the applicable tariff rate for the goods. The rate itself comes from the tariff schedule in force on the day; this course never teaches rates, only structure.

In Ghana, the customs value on which duty is computed is the CIF value, and Ghana Revenue Authority guidance explains that the subsequent import tax, VAT, is computed on the duty-inclusive value. CIF, cost, insurance and freight, is the Incoterms 2020 rule under which the

seller pays the costs, freight, and insurance to the named destination port; the next block treats these rules properly. The costing point here is that the valuation base itself is built from goods value, freight, and insurance, and each later assessment component has its own legally defined base stacked on the ones before it.

Now set the two concepts side by side. The customs value answers a legal question: on what value does the law compute duty for this consignment? The landed cost answers a business question: what did getting these goods to my warehouse, ready for sale, actually cost me? The landed cost contains the customs value's consequences, the duty and taxes computed on it, but also freight-side, destination-side, and tail costs the customs value never carries. The two figures coincide only by accident. Using one where the other is wanted produces two classic errors: pricing goods off the customs value, which understates the true cost, and filing the landed cost as the valuation base, which misstates the duty computation.

KEY CONCEPT

Incoterms 2020 and who pays which line

A landed cost is always built for a particular party, and the Incoterms 2020 rules decide which lines belong in that party's build. Incoterms 2020 is the International Chamber of Commerce's set of eleven three-letter rules allocating tasks, costs, and risk between seller and buyer in an international sale; they do not transfer ownership or fix payment terms. Lesson 4.3, Incoterms and Allocation of Responsibilities, taught the full allocation; this lesson applies the costing consequences.

Free on Board (FOB) is an Incoterms rule under which the seller delivers goods on board the vessel at the named port of shipment. Cost, Insurance and Freight (CIF) is an Incoterms rule under which the seller pays specified costs, freight, and insurance to the named destination port while risk transfers earlier at shipment.

Four costing consequences drive every build.

1. Main carriage. Under the E and F rules, including EXW and FOB, the buyer arranges and pays the main carriage, so the buyer's build carries a full freight line. Under the C and D rules, including CFR, CIF, CPT, CIP, DAP, DPU, and DDP, the seller pays carriage to the destination, so that cost reaches the buyer inside the goods price.

2. Risk travels separately from cost. Under the four C rules the seller pays carriage to the destination, but risk passes to the buyer already at shipment, on board the vessel at the port of shipment under CIF. A CIF buyer still carries the voyage risk and may insure in its own interest.

3. Insurance. Only CIF and CIP oblige the seller to insure: under CIF at minimum Institute Cargo Clauses (C) cover, under CIP at the higher (A) level. Under every other rule, insurance is the risk-bearing party's own arrangement.

4. Import clearance. Import formalities, duties, and taxes at destination are the buyer's side under all rules except DDP, where the seller carries them.

The practical rule for the build follows: every line the contract puts on the buyer appears in the buyer's landed cost, and every line it puts on the seller arrives already inside the goods price. A CIF Tema purchase needs no ocean-freight line in the buyer's build, but it still needs duty, destination charges, inland transport, and the tail.

KEY CONCEPT

Exchange rates and the moving build

An exchange rate is the rate at which one currency is exchanged for another. It is the quiet multiplier sitting on top of most of a landed-cost build, because international trade is invoiced across currencies: a Ghanaian importer typically buys goods in US dollars or euros, while selling them in Ghana cedis.

The arithmetic is direct. Every foreign-currency line in the build, the goods value above all, and commonly the ocean or air freight and some surcharges as well, is converted at the rate ruling when the money is actually paid. If the invoice is 10,000 US dollars, the cedi cost is 10,000 multiplied by whatever the rate is on payment day, not on order day. Between those two days the rate moves, and a movement that looks small per dollar becomes a large number across a consignment.

Three costing disciplines follow.

1. Date every conversion. A build should state which date's rate it used, because a build made at order time and a payment actually made weeks later can sit on different rates.

2. Re-run the build when the rate moves. A landed cost is a snapshot, not a permanent fact. Before reordering, and before setting a new selling price, the build is recomputed at the current rate.

3. Budget with a stated planning rate. Businesses that quote selling prices months ahead use a planning rate and treat the difference between it and the eventual rate as a risk to be managed, not as a surprise to be suffered.

Freight brings its own currency exposure. Fuel- and currency-related surcharges of the kind met in Lesson 7.4 move with the dollar side of shipping, so a weaker local currency can raise a freight quotation even when the underlying service is unchanged. Exchange-rate fluctuation is not a footnote to the landed cost; it touches several lines at once, and it can reverse a sourcing decision, as the second worked example shows.

KEY CONCEPT

Unit landed cost as the decision number

The end of every build is a division. Unit landed cost is the landed cost allocated to one unit of an imported product: the consignment's total landed cost divided by the number of sale units it contains, with the same care over the allocation basis as freight per unit required.

Unit landed cost is the number three kinds of decision run on.

1. Pricing. A selling price must recover the unit landed cost plus a margin. Pricing off the invoice price of the goods alone leaves freight, duty, destination charges, and the tail to be absorbed by money that was never there.

2. Sourcing and purchasing. Suppliers are compared on landed totals, not on ex-factory price tags. The second worked example reverses an apparently obvious choice precisely by running both suppliers through the full build.

3. Margin honesty. Management accounts that record goods at invoice value and discover the rest of the cost later make products look profitable until the freight bill, the assessment, and the storage invoices arrive. Recording inventory at landed cost keeps the profit calculation honest from the day the goods arrive.

Unit landed cost also behaves dynamically, and a costing team watches its drivers: the freight rate and the pricing basis behind it, the packing density that sets chargeable quantity, the exchange rate, the tariff treatment of the goods, and the tail charges that follow clearance speed. When any driver moves, the unit landed cost moves with it, which is why the build is kept as a live model and re-run rather than filed as a one-off calculation.

This closes the frame of the lesson: landed cost is the total, the cost breakdown is its anatomy, and unit landed cost is the decision number that pricing, purchasing, and margin reporting should all quote.

WORKED EXAMPLE

Building the landed cost of 500 cartons of fruit juice

Step 1: The consignment. Adwoa Foods Ltd, a distributor in Kumasi, imports 500 cartons of bottled fruit juice from an overseas supplier on FOB terms; the goods travel as an LCL consignment via Tema Port, then by truck to Kumasi. The invoice is 10,000 US dollars, worth GHS 120,000 at the assumed illustrative rate of 12 cedis to the dollar. All figures are illustrative.

Step 2: The freight line by weight or measurement. Each carton measures 0.4 by 0.3 by 0.3 metres and weighs 18 kilograms gross, so the consignment occupies 18 cubic metres and weighs 9 tonnes. Weight or measurement, whichever is greater: 18 beats 9, so the chargeable quantity is 18 revenue tonnes. At an illustrative GHS 600 per revenue tonne, ocean freight is 18 x 600 = GHS 10,800, which is GHS 21.60 inside every carton.

Step 3: Insurance. Cargo insurance for the sea leg: an illustrative premium of GHS 1,200.

Step 4: Duties, taxes, and levies. The customs value on the CIF basis is goods 120,000 plus freight 10,800 plus insurance 1,200 = GHS 132,000. Import duty is computed on that customs value at the applicable tariff rate, shown here as the illustrative assessment figure GHS 11,880. The import tax (VAT) and levies such as the NHIL, the GETFund levy, the ECOWAS levy, and the AU levy, each on its own legal base, are shown as a combined illustrative figure of GHS 24,460. These amounts are placeholders that teach structure only; the live figures are whatever the tariff and tax laws prescribe on the day of assessment.

Step 5: Destination and tail lines. Terminal handling and port charges at Tema: GHS 4,500. Inland transport to Kumasi: GHS 6,000. Clearance and agency fees: GHS 3,500. A storage and contingency allowance for the cost tail: GHS 2,000.

Step 6: Total and per unit. The build: goods 120,000; freight 10,800; insurance 1,200; duty 11,880; taxes and levies 24,460; terminal handling 4,500; inland transport 6,000; clearance and agency 3,500; allowance 2,000. Total landed cost: GHS 184,340. Unit landed cost: 184,340 divided by 500 = GHS 368.68 per carton. The goods contributed GHS 240 per carton; everything after the invoice added GHS 128.68, more than half as much again.

What does this example demonstrate? A landed cost is a build, run line by line in a fixed order: the freight line computed from a pricing basis, the duty lines on legally defined bases, the tail included. The per-carton figure is the only output a pricing decision can use.

Key lesson: price the consignment through the whole build, and label the placeholders for what they are: the live numbers come from the tariff and the market on the day.

WORKED EXAMPLE

When the cheaper supplier lands dearer

Step 1: The decision. Barimah Industrial Supplies, a distributor in Accra, needs 100 water pumps. Supplier A, a pump assembler based in Tema, quotes GHS 2,600 per pump delivered to Accra, inclusive of local taxes. Supplier B, a manufacturer in Shenzhen, quotes 15,000 US dollars for the hundred pumps on FOB Shenzhen terms: GHS 180,000 at the assumed illustrative rate of 12 cedis per dollar, or GHS 1,800 per pump. On price tags, Supplier B looks far cheaper. All figures are illustrative for teaching purposes.

Step 2: Supplier A's landed cost. The quote is delivered and tax-inclusive, so the build is short: goods GHS 260,000, plus receiving and handling at the Accra warehouse, an illustrative GHS 2,000. Total landed cost GHS 262,000, or GHS 2,620 per pump.

Step 3: Supplier B's freight line. Packed on pallets, the pumps measure 4 cubic metres in total and weigh 3.5 tonnes. Under weight or measurement, 4 beats 3.5, so the consignment pays on 4 revenue tonnes. At an illustrative GHS 900 per revenue tonne, ocean freight is 4 x 900 = GHS 3,600. Cargo insurance adds an illustrative GHS 1,500.

Step 4: Supplier B's duties and destination lines. The customs value on the CIF basis is 180,000 plus 3,600 plus 1,500 = GHS 185,100. Import duty at the applicable tariff rate: illustrative placeholder GHS 22,200. VAT and levies, each on its own legal base, combined: illustrative placeholder GHS 52,000. As before, the placeholders teach structure, not rates. Terminal handling and port charges: GHS 3,800. Clearance and agency fees: GHS 3,200. Inland transport from Tema to Accra: GHS 5,500. Storage and contingency allowance: GHS 1,500.

Step 5: Supplier B's total. 180,000 plus 3,600 plus 1,500 plus 22,200 plus 52,000 plus 3,800 plus 3,200 plus 5,500 plus 1,500 = GHS 273,300, which is GHS 2,733 per pump.

Step 6: The reversal. Supplier B won on the price tag by GHS 800 per pump and lost on landing by GHS 113, about GHS 11,300 across the order: the duty, tax, and destination lines outweighed the goods-price advantage. The comparison is also exposed to the exchange rate. At an illustrative 13 cedis to the dollar, Supplier B's goods alone cost about GHS 150 per pump more, roughly GHS 2,883 landed, before freight and surcharges that are also commonly dollar-linked. Supplier A's cedi price does not move.

What does this example demonstrate? An FOB price is one line of a build, not a total. The ranking of suppliers can reverse, and exchange-rate movement can widen the reversal.

Key lesson: compare source markets on landed totals computed at current rates, never on invoice prices, and re-run the comparison whenever rates or the exchange rate move.

WORKED EXAMPLE

Air or sea for urgent spare parts

Step 1: The problem. Naa Densua Machinery Ltd in Tamale maintains a customer's bottling line, which is down awaiting four cartons of spare parts from a supplier in Dubai, invoiced at 18,000 US dollars: GHS 216,000 at the illustrative rate of 12 cedis per dollar. The choice is air through Kotoka International Airport or sea through Tema Port; this example puts numbers on the air-versus-sea trade-off that Lesson 10.1 introduced qualitatively. All figures are illustrative.

Step 2: The air computation. Each carton measures 80 by 60 by 50 centimetres and weighs 30 kilograms gross. Volume per carton is 80 x 60 x 50 = 240,000 cubic centimetres; volumetric weight is 240,000 divided by 6,000 = 40 kilograms. Chargeable weight is the higher of the two, so the consignment pays on 4 x 40 = 160 chargeable kilograms. At an illustrative GHS 130 per chargeable kilogram, air freight is 160 x 130 = GHS 20,800, plus road delivery from Accra to Tamale of GHS 1,200: an air transport total of GHS 22,000. Illustrative door-to-door transit: 6 days.

Step 3: The sea computation. The same four cartons occupy 4 x 0.24 = 0.96 cubic metres and weigh 0.12 tonnes. Weight or measurement: 0.96 beats 0.12, so the consignment pays on 0.96 revenue tonnes. At an illustrative GHS 1,250 per revenue tonne, ocean freight is 0.96 x 1,250 = GHS 1,200, plus road from Tema to Tamale of GHS 2,800: a sea transport total of GHS 4,000. Illustrative door-to-door transit: 35 days.

Step 4: The carrying cost of waiting. At an illustrative holding cost of 20 per cent per year, the spares cost about 43,200 divided by 365 = about GHS 118 per day to carry. Sea's extra wait is 35 minus 6 = 29 days, about GHS 3,430 of extra carrying cost. Air's transport premium is 22,000 minus 4,000 = GHS 18,000, shrinking to roughly GHS 14,570 after the carrying saving.

Step 5: The downtime line. Each day the bottling line stands still costs the customer an illustrative GHS 6,000 of lost production. Sea's 29 extra days therefore cost about 174,000, dwarfing air's GHS 18,000 premium: for this emergency, air wins outright. But for the same spares ordered months ahead for a planned overhaul, the downtime line would disappear and sea's GHS 4,000 would beat air's GHS 22,000.

What does this example demonstrate? The freight difference is one line; time is another. Carrying cost, and where relevant the cost of a stopped operation, belong inside the comparison, and urgency is itself a cost line.

Key lesson: air or sea is a total-cost argument, not a freight-rate argument, and the same shipment can deserve opposite answers at different moments.

IMPORTANT NOTE

A landed-cost model is only as honest as its tail

The most common way a landed-cost calculation misleads is not an arithmetic error; it is an omission at the end. A build that stops at goods, freight, and duty looks respectable and still lies, because the buyer will really pay the lines that come after clearance: the clearing agent's fees, terminal handling, inland transport, storage while goods await collection, and the exposure to time-based charges if anything goes slowly. Lesson 9.4, Demurrage, Detention, and Storage Charges, treated what happens to a container that waits; the next lesson, Lesson 10.3, Storage, Detention, Demurrage, and Other Charges, prices the same family in detail. A build that pretends clearance is instant and storage is free has simply moved those costs off the page, not out of the world.

The discipline is twofold. First, include a realistic tail: the known fees as actual quotations, and an allowance for the unknown, as the worked examples did with their storage and contingency line. Second, state the exclusions next to the total. A figure labelled "landed cost, excluding storage beyond five days and excluding bank charges" is honest even if incomplete; an unqualified total invites decisions made on money that does not exist. When a colleague presents a landed cost, the professional question is not only "what did you include?" but "what did you leave out, and why?"

IMPORTANT NOTE

Duty and tax figures belong to the tariff law of the day

Every duty, tax, and levy figure in this lesson's examples is a labelled placeholder, and the label is the lesson. A template teaches the structure of the computation: goods, freight, and insurance forming the customs value; duty computed on that value at the applicable tariff rate;

each further tax and levy computed on its own legally defined base; the assessment generated for payment. The numbers that fill the structure are not teaching material at all. They are set by the tariff schedule and the tax laws in force on the day of assessment, and in Ghana they are computed when the declaration is assessed in ICUMS, with the tax bill paid at a participating bank.

Two cautions follow for any working model. First, never carry a placeholder into a live decision, and never assume yesterday's assessment repeats today's: rates, component lists, and bases change as laws change, and levies in particular are created and renamed over time. The authoritative sources are always the applicable tariff schedule and the revenue authority, the Ghana Revenue Authority in Ghana. Second, the same law-of-the-day caution applies to the freight side: the volumetric conversion factor used in an air computation is the one specified in the applicable tariff or contract, and carriers may publish factors other than the IATA standard. A landed-cost model is therefore a living template with empty cells for the day's numbers, never a finished price list.

IMPORTANT NOTE

Selling on ex-factory cost is how margins die

The recurring temptation this lesson exists to dismantle is pricing from the invoice. A trader buys at GHS 240 per carton, adds what looks like a healthy margin, and sells; the freight, the duty, the destination charges, and the tail then take GHS 128.68 per carton that the price never recovered, as the first worked example showed. The business has not priced its product; it has priced its supplier's product and donated the difference. Margin honesty begins with recording inventory at landed cost and quoting the unit landed cost whenever anyone asks what the goods cost.

The same discipline governs purchasing. The second worked example reversed a sourcing decision by GHS 113 per pump, and exchange-rate movement could widen that reversal without either supplier changing a price. A purchasing file that records only quotations records the smallest part of the decision; the file that matters keeps both builds, their dates, and the rates used.

Finally, keep the build alive. Unit landed cost moves when the freight market moves, when the exchange rate moves, when packing density changes, and when clearance slows. The costing habit that protects a business is to re-run the build before each reorder and each price review, so that the number on which pricing, purchasing, and margin reporting all rest is today's number, built today's way, with its tail included and its placeholders clearly marked.

Practice questions (Lesson Checks)

Practice Question 1

A freight forwarder in Tema receives an LCL consignment of generator parts imported from China for on-carriage to an inland customer: ten pallets measuring 6 cubic metres in total and

weighing 4,000 kilograms. The consolidator's tariff charges LCL freight on weight or measurement, whichever is greater.

Which statement correctly describes how the freight charge will be calculated?

1. On the gross weight of 4 tonnes, because ocean freight is always calculated on actual weight alone.

2. On 6 revenue tonnes, because the measurement of 6 cubic metres exceeds the weight of 4 tonnes under the weight-or-measurement rule. ✓

3. On 5 revenue tonnes, because the chargeable quantity is the average of the weight and the measurement.

4. On a full-container rate, because every consignment tendered to a consolidator is ultimately charged as a whole container.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Under the weight-or-measurement rule for LCL ocean freight, the consignment is charged on its weight in tonnes or its volume in cubic metres, whichever is greater, and the resulting chargeable unit is the revenue tonne. Here the consignment weighs 4 tonnes and measures 6 cubic metres; measurement is greater, so the charge is computed on 6 revenue tonnes. This mirrors the lesson's fruit-juice example, where 18 cubic metres beat 9 tonnes and set the chargeable quantity.

Answer 1 fails because weight alone is only half of the rule: a dense cargo would indeed pay by weight, but this consignment is bulkier than it is heavy. Answer 3 invents an averaging method that exists in no tariff; the rule takes the greater of the two figures precisely so the carrier is paid for whichever scarce resource, payload or space, the cargo actually consumes. Answer 4 confuses the physical consolidation, in which many LCL consignments share one container, with the charging method, under which each consignment pays its own share by revenue tonne rather than a whole-box rate.

The takeaway: before any freight figure enters a landed-cost build, read the pricing basis, because the basis, not the goods, decides the arithmetic.

Practice Question 2

After a costing review, a trainee tells her manager that the customs value shown on the import assessment is simply the landed cost of the shipment under another name, so the assessment figure can be used directly for setting selling prices.

Which statement best describes the relationship between customs value and landed cost?

1. The trainee is right: customs value and landed cost are two names for the same total, so the assessment can be used directly for pricing.

2. Customs value covers only the transport charges of a shipment, while landed cost covers only the duties and taxes, so together they add up to the total cost.

3. Customs value is the value determined for customs purposes under the applicable valuation rules, on which duty is computed at the applicable tariff rate, while landed cost is the wider business total that also carries freight-side, destination, and tail costs. ✓

4. Customs value is set by whatever figure the seller writes on the commercial invoice, while landed cost is set by the customs authority at the port of discharge.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The customs value answers a legal question: it is the value determined for customs purposes under the applicable valuation rules, starting in general from the transaction value, and duty is computed on it at the applicable tariff rate. The landed cost answers a business question: what did getting these goods to the warehouse, ready for sale, really cost? It contains the consequences of the customs value, the duty and taxes, but also insurance, terminal and handling charges, inland transport, clearance fees, and the cost tail. In Ghana the duty base is the CIF value and the assessment is generated in ICUMS, yet the landed total still carries lines the assessment never shows, which is why the fruit-juice build's landed cost of GHS 368.68 per carton far exceeded the value on which duty was computed.

Answer 1 is the trainee's error restated: the two figures coincide only by accident. Answer 2 swaps the contents of the two concepts entirely. Answer 4 is wrong on both halves: the customs value is determined by rule rather than by whatever the invoice asserts, and the customs authority does not set anyone's landed cost.

The takeaway: use the customs value for the duty computation and the unit landed cost for pricing, and never feed one to the other's decision.

Practice Question 3

An importer in Accra purchases a consignment on CIF Tema terms. A colleague preparing the landed-cost spreadsheet adds a line for ocean freight to Tema and another making the seller responsible for paying the import duty in Ghana.

Which option best explains the effect of the Incoterms rule on which cost lines belong in the buyer's build?

1. Both entries are correct: under CIF the buyer pays the ocean freight to Tema and the seller pays the import duty.

2. Under CIF the seller pays every charge up to the buyer's warehouse, including import duty, terminal handling, and inland transport, so the buyer's build needs no further lines after the goods price.

3. Under CIF the buyer pays no insurance and carries no risk, because the seller's responsibility continues until the goods are discharged at Tema.

4. Under CIF the seller pays freight and minimum insurance to the named destination port, so that cost reaches the buyer inside the goods price and the customs value, while import duty remains the buyer's side and the buyer still bears the transit risk from shipment. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

CIF, cost, insurance and freight, is an Incoterms 2020 rule under which the seller pays the specified costs, freight, and insurance to the named destination port while risk transfers earlier, when the goods are on board the vessel at the port of shipment. For the landed-cost build this means the ocean freight is not a separate buyer's line; it sits inside the goods price and, in Ghana, inside the CIF value on which duty is computed. Import clearance, duties, and taxes are the buyer's side under every rule except DDP, so the build still needs the duty and tax lines, the destination charges, inland transport, and the tail.

Answer 1 doubles the freight by paying it twice and wrongly hands import duty to the seller. Answer 2 describes something closer to DDP carried to the warehouse, which CIF is not; CIF stops at the named port and only for cost, never for risk. Answer 3 confuses cost with risk: under the C rules the seller pays carriage to destination, but the risk passes to the buyer already at shipment, and the seller's insurance obligation under CIF is only minimum cover for the buyer's benefit.

The takeaway: the Incoterms rule decides which lines move inside the goods price and which stay in the buyer's build; it never deletes them from the total.

Fact-check appendix (audit notes — not for entry)

Definitions of landed cost, freight costing, freight rate, chargeable weight, volumetric weight, customs value, import duty, import tax, cost breakdown, unit landed cost, CIF, and FOB follow the lesson brief's key terms, defined at first use and used naturally in the blocks. — source: lesson brief 10.2, Key Terms (curriculum database).

Weight-or-measurement (W/M) charging for LCL ocean freight, the revenue-tonne concept, and charging by container for FCL are general freight-industry pricing practice, taught as concepts with illustrative figures only. — source: general business knowledge, consistent with research sheet trade-transport-standards.md, section 7 (typical freight-rate components defined; no amounts taught).

IATA volumetric weight formula: L x W x H in centimetres divided by 6,000; chargeable weight is the higher of gross and volumetric weight; 6,000 is the IATA standard though some carriers or contracts use other divisors (5,000 or 7,000); one cubic metre equals about 166.67 kilograms under the standard factor. — source: research sheet trade-transport-standards.md, section 8.

Typical ocean freight-rate components named without amounts: base ocean freight, terminal handling charge (THC) at origin and destination, documentation fee, and security (ISPS) charge; bunker and currency adjustment factors exist as surcharge names. — source: research sheet trade-transport-standards.md, section 7.

Incoterms 2020: ICC's eleven three-letter rules allocating tasks, costs, and risk; they do not transfer ownership or fix payment terms; main carriage is the buyer's under E and F rules (EXW, FCA, FAS, FOB) and the seller's under C and D rules (CFR, CIF, CPT, CIP, DAP, DPU, DDP); under the four C rules the seller pays carriage to destination but risk passes at shipment; only CIF and CIP oblige the seller to insure (CIF minimum Institute Cargo Clauses (C), CIP the higher (A) level); import clearance, duties, and taxes are the buyer's under all rules except DDP. — source: research sheet trade-transport-standards.md, section 1 (safe phrasings).

FOB and CIF definitions as given follow the lesson brief's key terms, which match the research sheet's allocation of obligations. — source: lesson brief 10.2 Key Terms; research sheet trade-transport-standards.md, section 1.

WTO/WCO customs valuation: six methods applied sequentially, starting with transaction value (the price actually paid or payable for the goods when sold for export, with prescribed adjustments), taught as the general principle that duty is computed on a customs value determined by rule; no valuation figures taught. — source: research sheet trade-transport-standards.md, section 12.

In Ghana, an import assessment consists of import duty plus additional taxes and levies; the assessment is generated as a tax bill when the declaration is processed in ICUMS and is paid at a participating bank; the customs declaration is submitted in ICUMS by the clearing agent. — source: research sheet ghana-trade-customs.md, sections 1, 6, and 7 (safe phrasings).

Ghana assessment component names: import duty, VAT, NHIL (National Health Insurance Levy), GETFund levy, ECOWAS levy, and AU levy; GRA guidance explains that import duty is computed on the CIF value and VAT is computed on the duty-inclusive value. No rates, bands, or percentages are stated anywhere in the lesson. — source: verified 2026-10 against the Ghana Revenue Authority's own pages: https://gra.gov.gh/customs/vehicle-importation and https://gra.gov.gh/customs/customs-tariffs-and-levies (supplements research sheet ghana-trade-customs.md, section 7, which listed these names as unverified; verified here per WRITING_SPEC.md rule 5.2).

Freight cost per unit (spend divided by units shipped) is the standard freight comparator in performance measurement. — source: research sheet logistics-scm-frameworks.md, section 7.

Total-cost trade-off: slower, cheaper modes raise in-transit and cycle inventory cost; faster modes cost more per unit but cut inventory and stockout cost. — source: research sheet logistics-scm-frameworks.md, section 6.

Exchange-rate concept and the effect of currency movement on converted invoice values are general business knowledge, taught with an assumed illustrative rate only; no actual exchange rate is stated as fact. — source: general business knowledge; WRITING_SPEC.md rule 5.1.

All companies (Adwoa Foods Ltd, Barimah Industrial Supplies, Naa Densua Machinery Ltd), goods, quantities, dimensions, weights, freight rates, premiums, duty and tax placeholders, exchange rates, transit times, and holding-

cost percentages in the three worked examples and key concepts are illustrative for teaching purposes only; the duty and tax placeholders do not represent any actual tariff, tax, or levy rate or band, and real figures come from the tariff schedule, the tax laws, and published carrier tariffs on the day. — source: WRITING_SPEC.md fact-safety rule 5.1; the research sheets' DO NOT CLAIM lists.

Cross-references by name only, without re-teaching: Lesson 10.1 Components of Total Logistics Cost (cost families; air-versus-sea trade-off introduced qualitatively); Lesson 7.4 Shipping Charges, Freight Rates, and Surcharges (quotation assembly and surcharges); Lesson 7.3 Containerisation: FCL, LCL, and Container Types; Lesson 4.3 Incoterms and Allocation of Responsibilities; Lesson 9.4 Demurrage, Detention, and Storage Charges; Lesson 10.3 Storage, Detention, Demurrage, and Other Charges. — source: WRITING_SPEC.md section 7 (anti-repetition); lesson briefs for the named lessons.

Lesson 10.3 — Storage, Detention, Demurrage, and Other Charges

Summary

This lesson examines additional logistics expenses that arise from cargo storage, container usage, shipment delays, and operational services. Learners will explore how demurrage, detention, terminal storage, handling, and other charges are calculated. Emphasis is placed on identifying chargeable periods, interpreting tariffs, reviewing invoices, and preventing avoidable costs through effective shipment coordination.

Learning objectives

- Identify common additional charges associated with international shipments.
- Distinguish between demurrage, detention, and terminal storage charges.
- Explain how free-time allowances affect container-related costs.
- Interpret daily rates and progressive charging structures.
- Calculate charges using specified tariffs and chargeable periods.
- Identify circumstances that cause additional shipping and terminal expenses.
- Explain how delays in documentation and cargo collection affect logistics costs.
- Describe methods for reviewing and disputing incorrect charges.
- Recommend operational measures to minimise avoidable expenses.

Key terms

Demurrage — A charge commonly applied when a container remains within a terminal or designated facility beyond the applicable free-time allowance.

Detention — A charge commonly applied when a container is retained outside the terminal or designated facility beyond the permitted free-time allowance.

Storage Charge — A fee for keeping cargo or containers at a facility under the applicable tariff.

Free Time — A specified period during which particular container or storage charges are not applied.

Last Free Day — The final day of an applicable free-time allowance.

Chargeable Period — The period for which fees are calculated under an applicable tariff or agreement.

Progressive Tariff — A charging structure in which daily rates increase over successive periods.

Terminal Handling Charge (THC) — A charge associated with specified cargo or container handling services at a terminal.

Documentation Fee — A fee charged for preparing, issuing, or processing specified documents.

Container Cleaning Fee — A charge for cleaning a container when required under the applicable equipment-return conditions.

Container Repair Charge — A charge for repairing container damage where responsibility is established under the applicable agreement.

Invoice Dispute — A formal challenge concerning the accuracy or validity of a billed charge.

Cost Avoidance — Measures taken to prevent unnecessary expenditure before it occurs.

Teaching content

KEY CONCEPT

From the price of movement to the price of delay

The previous lesson, Freight Costing and Landed-Cost Calculations, answered the question every importer asks first: what will this shipment cost to bring in? It built the landed cost from freight, insurance, duties, taxes, and destination charges. That calculation prices the planned journey, the shipment behaving exactly as the quotation assumed.

This lesson prices a different thing: what the shipment can cost when the plan slips, together with the family of service charges that surrounds the movement itself. Lesson 10.1 mapped the components of total logistics cost and separated direct from indirect costs. Lesson 10.2 priced the plan. This lesson prices the tail: the charges that accrue at the end of the journey, after the vessel has berthed, when a container waits, a document fails, or equipment comes back in poor condition.

The mindset shift is the first learning. A buyer who asks only what the freight costs has priced the middle of the shipment and left both ends open. The professional question is broader: what can the tail of this shipment cost, which of those costs are within my control, and what would it cost to prevent them?

In costing language the family has two wings. The time-based charges, demurrage, detention, and storage, were defined and explained in Lesson 9.4, and this lesson refers back to those definitions rather than repeating them. Around them sits the wider family that this lesson's title calls other charges: documentation fees, cleaning and repair charges, weighing and

examination-related costs, and the internal costs a delay generates inside the importer's own business.

None of these are exotic. Every one of them appears on ordinary import invoices, and every one of them is either budgeted and managed, or unbudgeted and suffered.

KEY CONCEPT

The three time-based charges as budget lines

As the previous module taught, demurrage is a charge commonly applied when a container remains within a terminal or designated facility beyond the applicable free-time allowance; detention is a charge commonly applied when a container is retained outside the terminal beyond the permitted allowance; and a storage charge is a fee for keeping cargo or containers at a facility under the applicable tariff. Lesson 9.4 explained where each clock runs and when it starts. This lesson treats the same three as cost lines, and the costing view adds three distinctions the definitions alone do not show.

1. The creditor differs. Demurrage and detention are charged by the ocean carrier under the carrier's arrangement, while storage is charged by the terminal or facility under its own tariff. Two invoices from two organisations can arrive for the same container in the same week.

2. The resource priced differs. Demurrage prices a full container occupying terminal time, detention prices the carrier's equipment held out of circulation, and storage prices physical space. Because the resources differ, the charges can stack on the same days, which is why a delayed container costs more than newcomers expect.

3. The arithmetic differs from freight. Time-based charges are per container and per day, so the cost scales with time rather than with quantity or value, and under a progressive tariff the rate itself rises as the delay lengthens.

For budgeting, each of the three has the same expected value on a well-run shipment: zero. Each also carries the same tail risk: an escalating exposure that grows every day a process problem persists. That combination, expected zero with a heavy tail, is exactly the profile a budget must handle deliberately, and the rest of this lesson builds that handling.

KEY CONCEPT

Free time and the last free day as costing instruments

Free time is a specified period during which particular container or storage charges are not applied, and the last free day is the final day of that allowance. Lesson 9.4 explained how the allowance works; the costing lesson is what the allowance does to money.

Free time is the buffer that converts a potential charge into a zero. On the planned shipment, clearance, collection, and empty return all happen inside the allowance, so the time-based cost of the shipment is nil. The allowance is therefore not paperwork trivia. It is the risk margin built

into the shipment's cost structure: the number of days of operational slack the importer owns before delay starts converting directly into invoice value.

Two costing consequences follow. First, the expected cost of time-based charges on any shipment is zero if the plan holds, and the plan holds when the disciplines of earlier lessons hold: documents and permits complete and verified before arrival as Lessons 6.3 and 6.4 taught, the release chain worked early as Lesson 9.3 taught, and transport committed with margin. Second, the allowance's length sets how thin the margin is. A shorter allowance, which refrigerated equipment commonly carries, leaves less room for the ordinary frictions of clearance and collection, so the same slip produces charges on one shipment and none on another.

The last free day is the switch between two economies. Before it, an extra day of delay costs effort: a phone call, an overtime hour, a courier fee. After it, an extra day costs tariff money, at rates that commonly rise. Costing a shipment forward therefore means costing the effort that keeps it on the right side of that switch, which is the prevention economics this lesson builds later.

KEY CONCEPT

How the charges accrue and land as invoices

Lesson 9.4 explained that time-based charges accrue day by day and are invoiced after the event, with no visible meter at the gate. The costing lesson is what that behaviour does to a business's accounts, and it has four features.

1. Accrual is per container and per day. The unit of cost is one box for one billable day, and the chargeable period, meaning the period for which fees are calculated under the applicable tariff or agreement, runs from free-time expiry until collection for demurrage, from gate-out until empty return for detention, and for as long as cargo occupies charged space for storage. A container collected part-way through a day can still make the whole day billable, as Lesson 9.4 explained.

2. The invoice arrives after the money has already been committed by time. By the time it lands, the chargeable days are history. The finance team's first sight of the cost comes weeks after the operations team created it, which is why these charges feel like surprises even when the delay that caused them was well known inside the business.

3. The party invoiced is usually the consignee. Under Incoterms 2020, import clearance is the buyer's obligation under every rule except Delivered Duty Paid, so in most import transactions the party arranging clearance and collection is the party the clocks bill, as the previous module explained.

4. The amounts are reconstructed, not metered. A correct invoice is simply the tariff's arithmetic applied to the chargeable period, which means every time-based invoice can and should be

rebuilt from first principles: days, bands, rates. That reconstruction is the audit skill taught later in this lesson.

Seen this way, a demurrage invoice is not an anomaly. It is the accounting shadow of an operational event.

KEY CONCEPT

Interpreting daily rates and progressive tariffs

A progressive tariff is a charging structure in which daily rates increase over successive periods. Lesson 9.4 introduced the shape; this lesson develops the interpretation, because the difference between a flat rate and a stepped rate changes every forward-looking number in a logistics budget.

Interpreting a tariff for costing purposes means extracting four facts from it.

1. Where each band begins and ends. A stepped tariff might price the first chargeable days at one rate, the next block of days at a higher rate, and every day beyond at a higher rate still. The bands, not the headline rate, determine the cost of any given delay.

2. What each band costs per container per day. Because rates are stated per container per day, a multi-container shipment multiplies both the daily cost and the number of days the whole pattern runs.

3. How days are counted and from which event. The start event, the counting convention, and the part-day rule all come from the tariff, as Lesson 9.4 explained.

4. Whether the clocks run separately or in combination, because a combined arrangement changes which allowance protects which leg of the journey.

One fact-safe rule governs all of it: rates and free-time allowances are purely contractual terms of each line and terminal, and no standard amounts exist internationally. The only rates worth costing are the ones in the tariff attached to your own booking, and the only constant across tariffs is the shape of the behaviour: cost grows faster than time. When a tariff steps from one rate to a higher one, the marginal day is dearer than the average day, which is why a long delay devastates margins that a flat-rate estimate said were safe.

KEY CONCEPT

The wider family of other charges

The time-based three are the famous members of the family, but an import budget must expect a wider cast of other charges, most conditional on events that good operations make rare. The recurring ones are these.

1. Documentation fees. A fee charged for preparing, issuing, or processing specified documents, such as transport documents. A close cousin is the amendment fee, charged when a document

already issued must be corrected and reissued, which is a defect fee in disguise: it prices an error, not a service.

2. Terminal handling charges. A terminal handling charge is a charge associated with specified cargo or container handling services at a terminal, typically settled around release among the applicable payments Lesson 9.3 described. It is planned, unlike the clock charges, and belongs in every landed-cost calculation as Lesson 10.2 taught.

3. Container cleaning fees. A charge for cleaning a container when its return condition requires cleaning under the applicable equipment-return conditions. Residue, staining, odour, or pest traces can trigger it.

4. Container repair charges. A charge for repairing container damage where responsibility is established under the applicable agreement. The qualifier matters: responsibility must be established, and the equipment interchange record from Lesson 9.3 is usually where it is established or refuted.

5. Weighing fees. Where a shipment or container must be weighed, for example when a verified gross mass is established by weighing the packed container, the facility may charge for the service.

6. Inspection-related costs. When customs selects a consignment for examination, the unstacking, handling, and re-presentation of cargo take labour and time, facilities may charge under their tariffs, and the importer's agent spends hours in attendance.

7. Delay-linked internal costs. Re-lodgement of a corrected declaration, agency overtime, rebooked trucks, warehouse rescheduling, and the staff hours in between. They never appear on an external tariff, but they are real money, and Lesson 9.4 named them the standstill costs.

8. Storage beyond the port. The storage charge concept is not confined to terminals; cargo moved into a bonded or private warehouse sits under that facility's tariff instead.

9. Insurance excesses on claims. Where damage becomes an insurance claim, the policy's excess, the portion of each claim the insured bears, costs money even when the claim succeeds.

The family shares one character: every line is conditional on an event, and every event is one that planning can usually prevent.

KEY CONCEPT

How delays in documents and collection become cost

The clearest way to see why documentation delays and slow collection cost money is to trace the cascade, because the clock charges are only its most visible stage.

Consider a documentation delay. A required permit is not in hand when the vessel berths, or a commercial invoice disagrees with the transport document, and clearance stalls, as Lessons 6.3

and 6.4 each explained in their own territory. The stages that follow are these: the chargeable period opens and demurrage and terminal storage begin; the agent's time goes into corrections, re-lodgement, and follow-up; amendment fees may be incurred to fix documents already issued; the booked truck misses its slot and is rebooked; the goods reach the market later than the sales plan assumed; and, if the delay is long, private warehouse storage or an insurance claim may follow. One missing document, many line items.

Consider a collection delay. Release is complete but no truck is committed, or the warehouse is full, or stripping drags on. Demurrage runs until gate-out, then detention runs until the empty box is returned, and the terminal's space charge can run alongside the first of them. The cascade here is shorter, but the arithmetic is identical: a gap between two operational events is filled with tariff money.

The advanced insight is that the invoice understates the true cost. The external charges are receipted and visible; the internal costs, staff hours, missed sales, credibility with customers, are mostly invisible and never invoiced by anyone. Businesses that cost delays only by their invoices systematically understate what poor coordination costs them, and therefore underinvest in prevention. Once the whole cascade is priced, the case for spending on documents, permits, and transport booking before arrival becomes overwhelming.

KEY CONCEPT

Costing forward: the expected-charges budget

Everything so far explains how the charges behave after the event. The professional discipline is to cost them forward, before the vessel sails, shipment by shipment.

The tool is an expected-charges budget built for each shipment alongside the landed-cost calculation of Lesson 10.2. It lists every charge the shipment can expect with its source: ocean freight and surcharges from the quote, terminal handling and documentation fees from the tariff, duties and taxes from the assessment basis, transport from the haulier's rates. The time-based charges are budgeted at expected zero, because on a shipment whose documents, permits, release chain, and transport are all prepared, zero is genuinely the expected outcome. Expected zero is not wishful thinking; it is a prediction with a process behind it.

Alongside the expected lines sits a contingency understanding. A reserve is held for the tail, sized from the importer's own history of delays, and its purpose is to absorb a genuine shock, a customs examination, a storm-delayed vessel, without destabilising the month's accounts. The contingency is management's acknowledgment that plans meet reality; it is not a licence to be casual about the last free day, and a contingency that is regularly consumed is a signal that the underlying processes, not the budget, need work.

The discipline completes with tracking. Actual charges are recorded against the budget for every shipment, and variances are investigated with the same seriousness as a stock discrepancy: a cleaning fee, a detention charge, or a demurrage invoice is a variance with a cause, and the

cause sits in a process. This is cost avoidance, the practice of preventing unnecessary expenditure before it occurs, and it differs from cost negotiation, which happens after the expenditure exists and recovers only part of it.

KEY CONCEPT

Every demurrage invoice is a process failure

When a time-based charge accrues despite the expected-zero budget, the costing response is neither to pay and forget nor simply to argue the invoice down. It is to ask what process failed. Handled this way, every demurrage invoice becomes diagnostic: it tells the business exactly which discipline broke.

The recurring root causes are few, and each belongs to an earlier lesson.

1. Permits and certificates not obtained in time, so clearance cannot complete. Lesson 6.3 covered these documents; the failure sits in the pre-arrival document plan.

2. Document discrepancies discovered at lodgement, so release stalls while papers are corrected. Lesson 6.4 covered verification; the failure is in checking before arrival rather than after.

3. The release chain left incomplete, with shipping line release, customs release, or terminal release still outstanding while free time drains. Lesson 9.3 covered the three keys; the failure is in starting them early.

4. Transport not booked, or booked without margin, so a released container waits for a truck. The failure is in collection planning.

5. Communication failures, where a foreseeable delay was not escalated early enough to be managed. Lesson 5.4 covered tracking and communication; the failure is in escalation.

The working tool is a root-cause table with one row per invoice: the shipment, the chargeable days, the immediate cause, the underlying process failure, the owner of that process, the corrective action taken, and the date the fix was verified as working. The table converts invoices into an improvement pipeline.

Two rules make it work. First, fix the process, not the invoice: a negotiated discount leaves the failure in place to bill again next month. Second, verify the fix: if the same root cause produces a second invoice, the corrective action did not work and must be redesigned. The worked example later in this lesson shows one importer's quarterly table in full.

KEY CONCEPT

The economics of prevention and the KPI of zero

Prevention has a price: an agent working overtime, a courier carrying a certificate, a permit application lodged weeks early, a phone call to confirm a truck. Delay has a price too:

chargeable days at rates that rise in steps, stacked with storage and standstill costs. The economics of this lesson rest on comparing the two prices honestly, and the comparison is rarely close. Prevention buys a bounded, one-off effort cost, typically one day of expedited work. Delay buys an open-ended, escalating tariff cost that continues until someone acts. The worked example that follows puts illustrative numbers on exactly this comparison, and the pattern holds whenever the comparison is made seriously: prevention is almost always cheaper, often by several times.

Two management practices convert that economics into routine.

First, act inside the window where action is cheap. Time-based charges are negotiable only before they accrue: a request for an extension of free time, made early with a credible reason, is a routine commercial request that lines handle every day; the same request after the charges have accrued is an argument about money already owed. The cheap day is the day before the last free day, not the day the invoice arrives.

Second, measure the tail. The key performance indicator for this cost family is time-based and other avoidable charges as a percentage of total logistics spend, tracked period by period. No universal benchmark exists, and none is needed: the comparison is with your own past quarters. The goal is a trend toward zero, because zero is what a well-run import operation should accrue. Track the count of invoices as well as their value; a falling total with recurring invoices means a few processes still fail, and the root-cause table names them.

KEY CONCEPT

Reviewing and disputing incorrect charges

Because these charges are reconstructed from tariffs rather than metered, some of them are simply wrong: days miscounted, rates applied from the wrong band, a container billed after it was returned, damage charged that predates the consignee's custody. The review discipline is therefore not optional scepticism but routine audit: every time-based invoice and every equipment-related charge is checked against the tariff and records you already hold before it is paid. An accepted invoice is accepted practice; pay whatever arrives unexamined and you have certified its accuracy for every shipment that follows.

The audit reconstructs the charge from first principles. For a time-based charge: the container number matches your shipment; the availability, gate-out, and return dates match your records; the chargeable days are counted correctly under the tariff's convention; each day is priced from the correct band of the progressive structure; the arithmetic totals; and charges from different creditors correspond to their own tariffs and allowances, since the line's demurrage and the terminal's storage are separate charges that can legitimately both apply. For equipment charges: the cleaning or repair is described specifically enough to verify, and the responsibility question is answered by the equipment interchange record from Lesson 9.3, which recorded the container's condition at each interchange.

Where the reconstruction fails, the remedy is an invoice dispute, a formal challenge concerning the accuracy or validity of a billed charge. A dispute is conducted in writing, raised promptly, and supported by the evidence trail: the booking and its tariff schedule, the reconstruction, the interchange records, and photographs taken at unpacking and at empty return. The worked example later in this lesson shows one charge paid because the evidence supported it and another withdrawn because the evidence contradicted it, which is exactly the intended outcome of a disciplined review: valid charges are paid promptly, and invalid ones are not paid at all.

WORKED EXAMPLE

The arithmetic of one late container at Tema

A distributor of packaged food products in Accra is importing one forty-foot container of cooking oil through the Port of Tema. All figures and periods in this example are illustrative for teaching purposes.

Step 1: The booking terms. The booking confirmation carries an illustrative free-time allowance of five days for collection, counted from the day the container is made available, and a progressive demurrage tariff: for illustration, chargeable days one to three at GHS 300 per day, days four to six at GHS 500 per day, and every day from the seventh at GHS 800 per day. The terminal's illustrative storage charge is GHS 150 per day. The distributor's expected-charges budget carries demurrage and storage at zero.

Step 2: The failure. The goods require an import permit, and the application was never lodged before arrival, repeating the exact failure Lesson 6.3 warns against. Clearance cannot complete, and the container is not collected until the eighth chargeable day.

Step 3: The accrual table. Reconstructing the demurrage:

- Chargeable days one to three: three days at GHS 300, which is GHS 900.
- Chargeable days four to six: three days at GHS 500, which is GHS 1,500.
- Chargeable days seven and eight: two days at GHS 800, which is GHS 1,600.
- Demurrage total: GHS 900 plus GHS 1,500 plus GHS 1,600, which is GHS 4,000.

Terminal storage runs for the same eight days at GHS 150 per day, adding GHS 1,200. The two invoices together total GHS 5,200, and the rebooked truck adds an illustrative GHS 400.

Step 4: The counterfactual. Suppose instead that, in the week before arrival, the company had paid its clearing agent for one day of overtime at an illustrative GHS 600 and an illustrative GHS 250 courier fee to obtain the outstanding supporting document in time. Clearance completes inside free time, the container gates out on day four, and every time-based charge is zero. Total prevention cost: GHS 850.

Step 5: The comparison. GHS 5,200 against GHS 850: the delay cost roughly six times the prevention, and the GHS 4,350 difference bought nothing but days. The goods also reached the market eight working days later, which cost sales that nobody invoiced for.

What does this example demonstrate? The escalation pattern makes each late day dearer than the one before, so a delay of eight chargeable days costs far more than twice a delay of four. The invoice records only the external half of the loss. And the prevention side of the ledger is small, bounded, and known in advance.

Key lesson: when one day of expedited effort is compared honestly with one week of escalating charges, prevention is not a virtue. It is arithmetic.

WORKED EXAMPLE

A quarterly review: three invoices, three root causes

Kwabena Trading Ltd is a distributor importing consumer goods through the Port of Tema, its customs declarations lodged in ICUMS by its clearing agent, as Ghanaian import practice generally runs. In one quarter the company received three demurrage invoices, and its new logistics manager did what this lesson prescribes: treated each invoice as a process failure and built the root-cause table. All figures in this example are illustrative for teaching purposes.

Invoice one, an illustrative GHS 2,350 on a consignment of food products requiring a permit from the Food and Drugs Authority. Immediate cause: clearance could not complete. Root cause, traced with the agent: the permit application had not been lodged before the vessel sailed from its last port, so the authority's processing time was consumed after arrival instead of before it. Owner: the import officer responsible for regulatory documents. Corrective action: every permit application lodged at booking confirmation, with the expected approval tracked before the vessel's arrival, as Lesson 6.3 implies.

Invoice two, an illustrative GHS 1,050 on a consignment of household goods. Immediate cause: the commercial invoice's goods description did not match the transport document, and the declaration required correction. Root cause: documents were compiled in the arrival week instead of verified before it, the habit Lesson 6.4 teaches against. Owner: the documentation clerk. Corrective action: a pre-arrival verification checklist, with the agent confirming the consistency of descriptions, quantities, and references before lodgement.

Invoice three, an illustrative GHS 1,500 on a machinery consignment. Immediate cause: release was complete, but the first available truck came four days later. Root cause: transport was booked after release instead of against the planned release date. Owner: the transport coordinator. Corrective action: haulage committed at the same time as the release chain starts, a backup haulier identified, and the last free day diarised as a hard deadline, as Lesson 9.4 recommended.

The quarter's total, an illustrative GHS 4,900, matters less than the table's last column: three different owners, three different processes, and no common villain. The line, the terminal, and customs appear nowhere in the root causes. In the following quarter, with the three fixes implemented and verified, the company recorded no time-based charges on its Tema arrivals.

What does this example demonstrate? Demurrage invoices look like port costs and are usually process costs. A root-cause table converts a bad quarter into a work list, assigns each item to an owner, and makes each fix verifiable.

Key lesson: the review that pays for itself is not the one that argues with the invoices. It is the one that finds the three processes behind them and closes each one.

WORKED EXAMPLE

Cleaning and damage charges in Singapore: paying with evidence, disputing with evidence

An importer of industrial spare parts in Singapore collects a container from the terminal, unpacks it at its warehouse, and returns the empty box to the line's depot within the allowance. Two weeks later an invoice arrives with two lines: a container cleaning fee and a container repair charge. All figures in this example are illustrative for teaching purposes.

Step 1: Read the lines against the definitions. A cleaning fee is chargeable when the return condition requires cleaning under the equipment-return conditions, and a repair charge where responsibility is established under the applicable agreement. The audit question is the same for each line: what event produced this charge, and does the evidence support that event?

Step 2: The cleaning charge. The line's photographs show dried cargo residue on the container floor: several cartons of machine lubricant leaked during the voyage and in the warehouse. The importer's own unpacking report, compiled the day the container was stripped, records the same leakage. The evidence supports the charge, so an illustrative US dollars 180 cleaning fee is paid without dispute. The deeper lesson goes into the root-cause table: the leak was a packaging failure, and the corrective action is an improved packing specification on the next order, which is cost avoidance applied to a charge the importer would otherwise pay again.

Step 3: The repair charge. The line claims an illustrative US dollars 420 for a dented door panel and a cut door seal, implying damage in the importer's custody. The importer checks its copy of the equipment interchange record, the document Lesson 9.3 introduced as the record of condition at each interchange. The record signed at gate-out from the terminal already notes the dent and the cut seal as existing damage at the moment of collection. Damage that exists at collection cannot have been caused after it.

Step 4: The dispute. The importer writes to the line promptly, encloses the interchange record and the unpacking photographs, and states that responsibility for the damage is not established. The line checks its own interchange records, confirms the damage was pre-existing, and withdraws the charge.

What does this example demonstrate? The audit found one valid charge and one invalid one, and the evidence decided both. Paying the valid charge promptly preserved credibility; disputing the invalid one with documents recovered US dollars 420 that blind payment would have donated. Paying the invoice unexamined would have taught the opposite lesson to everyone, including the importer's own team.

Key lesson: every charge line is a claim until the evidence confirms it, and the evidence, the tariff, the unpacking report, the interchange record, the photographs, is assembled during the shipment, not after the invoice arrives.

IMPORTANT NOTE

Extensions are routine before the clock starts; arguments after it stops

Free-time allowances feel fixed, but they sit inside commercial relationships, and lines extend them for credible operational reasons more often than newcomers expect. The timing is everything. A request made days before the last free day, explaining the delay, proposing a realistic new collection date, and backed by a shipment history of prompt collections, is a routine commercial conversation. The same request made after the charges have accrued is something quite different: an argument about money the tariff already says is owed, conducted with a creditor who holds the records and owes the importer nothing.

This asymmetry has a budgeting consequence. The cheap time to solve a foreseeable delay costs a phone call; the expensive time costs the invoice. So the moment a delay becomes foreseeable, a permit running late, an examination notified, a truck unavailable, the escalation should go to the line's agent immediately, as Lesson 5.4's communication discipline implies, while the request still costs nothing.

The mirror-image rule applies to the invoices they send you. An accepted invoice is accepted practice: pay a time-based invoice without checking it against the tariff you were given, and you have verified nothing, but you have certified everything. Audit first, pay promptly what survives the audit, and dispute what does not.

IMPORTANT NOTE

Budget for the tail or the tail will budget for you

The landed-cost calculation of Lesson 10.2 prices the plan, and a budget built only from landed costs has a hole exactly the shape of this lesson. The hole closes in one of two ways. Either the importer prices the tail deliberately, an expected-zero line for the time-based charges, planned lines for handling and documentation, and a contingency reserve sized from its own delay history, or the tail prices itself, arriving as invoices nobody expected and taking its share from margins nobody protected.

Unpriced tail costs do their worst damage quietly. They inflate the true landed cost of the affected shipments above the figure used to set prices. They arrive in months whose budgets have already been spent. And because they are infrequent, they evade the per-unit costing routines that catch ordinary cost creep, so a business can run for years treating demurrage as bad luck before anyone adds up what the luck costs.

The discipline is not complicated. Add the tail lines to every shipment budget. Record actuals against them. Feed every variance into the root-cause table. Track the family's share of logistics spend over time and aim its trend at zero. None of this prevents a genuine shock, but it converts

the tail from a surprise into a managed, measured, shrinking category, which is the whole difference between a cost you suffer and a cost you run.

IMPORTANT NOTE

A dispute is an evidence exercise, conducted promptly and in writing

Disputing a charge is not a negotiation of mood or volume. It is an evidence exercise, and it has a method.

State the challenge in writing, promptly, while records are fresh and before the claim ages. Identify the precise defect: the days miscounted, the band misapplied, the container returned before the billed date, the damage that predates your custody. Attach the reconstruction and the documents: the booking and its tariff schedule, the equipment interchange record from each interchange, the unpacking report, and the photographs taken at collection and at return. Ask for the specific remedy: a corrected invoice or a withdrawal.

Keep the dispute narrow. Where an invoice carries five lines and two are wrong, dispute the two and pay the three, saying so in the letter. Withholding an entire account over one disputed line damages credibility and, with time-based charges that lines commonly require to be settled before release, can stall a container and breed exactly the charges being disputed. Credibility is the disputing party's main asset: the importer who audits everything, pays what survives, and disputes with documents is believed; the importer who disputes everything is filtered out.

One boundary completes the discipline. A dispute challenges accuracy and validity; it does not manufacture sympathy. The evidence either supports the charge or it does not, and a business that has kept the records knows which before the letter is written.

Practice questions (Lesson Checks)

Practice Question 1

An importer's booking at Tema carries an illustrative progressive demurrage tariff: chargeable days one to three at GHS 200 per day, days four to six at GHS 400 per day, and every day from the seventh at GHS 700 per day. The container is collected on the fifth chargeable day.

What demurrage charge does the tariff produce?

1. GHS 1,000, because all five days are priced at the first band's rate.

2. GHS 1,400, because the first three days are priced at the first band and the remaining two days at the second band. ✓

3. GHS 2,000, because all five days are priced at the second band's rate.

4. GHS 3,500, because once the highest band is reached its rate applies to the whole chargeable period.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

A progressive tariff prices each chargeable day from the band that day falls in, so the calculation is built band by band. Days one to three sit in the first band: three days at GHS 200, which is GHS 600. Days four and five sit in the second band: two days at GHS 400, which is GHS 800. The total is GHS 600 plus GHS 800, or GHS 1,400. The third band is never reached, because the container was collected on the fifth chargeable day.

Answer 1 applies the first band's rate to days that have already moved into the second band. Answer 3 applies the second band's rate to all five days, including the three that belong to the first. Answer 4 describes a structure in which the highest reached rate applies retrospectively to every day; that is not how stepped progressive tariffs are read, and it is precisely the misreading that a band-by-band reconstruction prevents.

The takeaway: reconstruct every time-based charge from the tariff, band by band and day by day, before accepting the figure.

Practice Question 2

Over one quarter, an importer receives three demurrage invoices on three different Tema arrivals, and each charge is valid under the applicable tariffs. At the quarterly review, which response best reflects the discipline this lesson teaches?

1. Negotiate a standing demurrage discount with the shipping line so that future invoices are smaller.

2. Increase next quarter's demurrage budget line so that the charges are covered in advance.

3. Trace each invoice to its process root cause, assign an owner to each cause, implement corrective actions, and track time-based charges as a share of logistics spend with the goal of trending to zero. ✓

4. Accept the charges as an unavoidable cost of shipping through busy ports and spread them across all shipments in the pricing.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The lesson's central discipline treats every demurrage invoice as a process failure with a discoverable root cause: a permit lodged late, a document discrepancy, transport not booked with margin. The root-cause table converts the invoices into an improvement pipeline with owners and verified fixes, and the key performance indicator, time-based charges as a share of logistics spend, trending toward zero, measures whether those fixes work. On a well-run shipment the expected time-based charge is zero, so zero is the legitimate target.

Answer 1 negotiates the symptom and leaves the failures in place to bill again next quarter. Answer 2 institutionalises the cost: budgeting more for demurrage is planning to keep failing, when the expected line should be zero with a contingency reserve reserved for genuine shocks. Answer 4 surrenders control, treating avoidable process costs as fate; valid charges are payable once, but their causes are fixable, and a business that prices them into every shipment taxes its good shipments for its bad processes.

The takeaway: the charges are valid, so the argument is not with the invoices. It is with the processes that produced them.

Practice Question 3

An importer receives an invoice from a shipping line containing a container cleaning fee, supported by photographs of cargo residue from cartons that leaked inside the box, and a container repair charge for door damage. The importer's copy of the equipment interchange record signed at collection records the same door damage as already existing when the container was handed over.

Which statement best describes the correct handling of the two charges?

1. Both charges should be paid, because an invoice issued by a shipping line is presumed accurate.

2. Both charges should be refused, because charges arriving after the cargo has been delivered cannot be valid.

3. The cleaning fee should be paid because the evidence supports it, while the repair charge should be disputed promptly in writing with the interchange record and photographs, because the damage predated the importer's custody. ✓

4. The repair charge should be paid now and disputed later if similar charges recur, while the cleaning fee should be passed to the supplier of the goods for immediate payment.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

A container repair charge is claimable where responsibility for the damage is established under the applicable agreement. The equipment interchange record from Lesson 9.3 is the document that records a container's condition at each interchange, and its note of the damage at collection places the dent before the importer's custody, so responsibility for it is not established. A prompt, written dispute with the record and photographs attached is the correct remedy. The cleaning fee is the opposite case: the residue is cargo-derived, and the photographs and unpacking report support it, so it is a valid cost of the shipment's own packing failure, to be paid and then prevented through a better packing specification next time.

Answer 1 is the blind payment this lesson warns against: an accepted invoice becomes accepted practice. Answer 2 refuses valid and invalid charges alike, sacrificing credibility and ignoring the evidence. Answer 4 delays a dispute it could win now, and misdirects the cleaning fee, which is the importer's own packaging lesson to pay for and fix.

The takeaway: audit every charge line against the evidence, pay promptly what the evidence supports, and dispute in writing what it contradicts.

Fact-check appendix (audit notes — not for entry)

Demurrage is a charge commonly applied when a container remains within a terminal or designated facility beyond the applicable free-time allowance; detention is a charge commonly applied when a container is retained outside the terminal beyond the permitted allowance; a storage charge is a fee for keeping cargo or containers at a facility under the applicable tariff — source: lesson brief key terms and research sheet trade-transport-standards.md section 11 (demurrage/detention charged by the ocean carrier, storage by the terminal, separate and additional; iContainers, https://www.icontainers.com/help/demurrage-detention-per-diem-shipping-delay-charges; Maersk explainer)

Free time is the allowance before time-based charges begin and the last free day is its final day; rates and free days are purely contractual tariffs of each line and terminal with no standard international amounts — source: research sheet section 11 and its DO NOT CLAIM item 5 (no amounts or day counts stated as fact in this lesson; all worked-example figures are labelled illustrative per writing spec rule 5.1)

Progressive tariff, chargeable period, THC, documentation fee, container cleaning fee, container repair charge, invoice dispute, cost avoidance are taught as defined in the lesson brief's key terms — source: curriculum lesson brief module-10/lesson-10.3.md

THC and documentation fee are standard, universally used charge names whose application varies by line and trade — source: research sheet section 7, typical freight-rate components (confidence medium-high; no amounts taught)

A verified gross mass may be established by weighing the packed container (SOLAS VI/2, Method 1), which is the basis for teaching weighing fees generically as a facility service — source: research sheet section 7 (IMO, https://www.imo.org/en/OurWork/Safety/Pages/Verification-of-the-gross-mass.aspx)

Customs risk management channels consignments to examination, which supports the generic treatment of inspection-related costs (no amounts, no Ghana-specific fees) — source: research sheet section 12

Incoterms 2020: import clearance is the buyer's obligation under all rules except DDP, so the consignee is typically the party invoiced for destination-side charges — source: research sheet section 1 (ICC, https://iccwbo.org/business-solutions/incoterms-rules/incoterms-2020/)

The equipment interchange record acknowledges the transfer of a container and records relevant condition details, making it the decisive evidence in cleaning and repair disputes — source: curriculum lesson brief module-09/lesson-9.3.md (EIR key term)

In Ghana, the customs declaration is submitted in ICUMS by the clearing agent; imports of regulated goods may require permits from agencies such as the Food and Drugs Authority — source: research sheet ghana-trade-customs.md sections 1 and 5

Insurance excess (the portion of each claim borne by the insured), amendment fees for corrected documents, re-lodgement of corrected declarations, early extension requests being routine while post-accrual requests are arguments, the audit-before-payment discipline, and the KPI concept of tracking a cost family as a percentage of spend over time are general business knowledge — source: general business practice, consistent with research sheet logistics-scm-frameworks.md section 7 (KPI cost-ratio concepts, no benchmarks)

Cross-references: Lesson 5.4 Shipment Tracking and Customer Communication; Lesson 6.3 Certificates of Origin, Permits, and Supporting Documents; Lesson 6.4 Document Verification and Common Discrepancies; Lesson 9.3 Cargo Release, Delivery Orders, and Terminal Procedures; Lesson 9.4 Demurrage, Detention, and Storage Charges; Lesson 10.1 Components of Total Logistics Cost; Lesson 10.2 Freight Costing and Landed-Cost Calculations — source: curriculum lesson briefs

Lesson 10.4 — Preparing and Evaluating Logistics Quotations

Summary

This lesson develops learners' ability to prepare professional logistics quotations and assess competing service offers. Learners will examine customer requirements, service scope, cost calculations, pricing margins, exclusions, quotation validity, and contractual conditions. The lesson also explains how to compare quotations based on total cost, reliability, transit time, service coverage, and commercial risk.

Learning objectives

- Define a logistics quotation and explain its commercial purpose.
- Identify the information required to prepare an accurate quotation.
- Distinguish between cost price, selling price, markup, and profit margin.
- Calculate a proposed selling price using specified costs and pricing rules.
- Identify the essential components of a professional freight quotation.
- Explain the importance of clearly stating inclusions, exclusions, and validity periods.
- Compare competing quotations using financial and operational criteria.
- Recognise common pricing errors and hidden-cost risks.
- Prepare a basic logistics quotation for a specified shipment.

Key terms

Logistics Quotation — A commercial offer stating proposed charges and conditions for specified logistics services.

Cost Price — The amount incurred in obtaining or delivering a service before adding a selling markup.

Selling Price — The amount charged to a customer for a product or service.

Markup — The amount added to cost to determine a selling price, commonly expressed as a percentage of cost.

Profit Margin — The proportion of selling revenue remaining after specified costs are deducted.

Service Scope — The activities and responsibilities included in a proposed service.

Quotation Validity — The period during which a commercial offer remains effective under its stated conditions.

Inclusions — Services and charges expressly covered by a quotation.

Exclusions — Services, charges, or responsibilities not covered by a quotation.

Rate Comparison — The evaluation of competing prices and commercial conditions.

Service-Level Agreement (SLA) — An agreement defining specified service expectations, performance measures, and responsibilities.

Commercial Terms — The conditions governing pricing, payment, service delivery, and related obligations.

Quotation Evaluation — The assessment of a commercial offer against cost, service, risk, and operational requirements.

Teaching content

KEY CONCEPT

What a logistics quotation is and what it is for

A logistics quotation is a commercial offer stating proposed charges and conditions for specified logistics services. When a business asks three freight forwarders what it would cost to move four pallets of goods from Accra to Antwerp, the documents those forwarders return are quotations: each states what the service will cost, what the service covers, and on what conditions the price stands.

The quotation serves both sides of the transaction. For the buyer, it converts an enquiry into a price that can be compared, budgeted, and approved internally. For the provider, it is the moment at which the business commits itself: once the customer accepts the quotation on its terms, the provider is expected to perform the stated scope at the stated price. The quotation therefore has a commercial purpose beyond information. It is the document on which the booking is based, the standard against which later invoices should be checked, and the boundary between a smooth service relationship and a string of disputes.

It is equally important to know what a quotation is not. A quotation is an offer, not a contract of carriage. It becomes commercially binding through acceptance on its terms, within the period for which it is offered; the contract of carriage itself arises later, through the booking and the transport document. That distinction, examined in the closing notes of this lesson, drives much of the professional discipline taught here.

The lesson has two halves. First, preparing a quotation: the costing discipline by which a provider estimates each cost family, adds the commercial layer, and states scope, validity, and conditions. Second, evaluating a quotation: the professional checklist by which a buyer compares competing offers on total cost and commercial risk rather than on headline price.

KEY CONCEPT

The information required to prepare an accurate quotation

Lesson 5.2, Freight Quotations and Shipment Booking, examined the enquiry from the shipper's side: what the buyer must tell the forwarder so that a meaningful price can be returned. This lesson picks up the same document from the other side of the desk, where the costing professional receives the enquiry, tests it, and prices it. The information required to prepare an accurate quotation covers the following.

1. The cargo. Description, number of packages or pallets, gross weight, dimensions, and packaging type. Vague descriptions prevent accurate costing; the professional confirms what the cargo actually is, because weight, volume, handling requirements, and regulatory treatment all hang on it. Lesson 5.2's rule that chargeable weight is priced on the greater of actual weight and volumetric weight is the arithmetic expression of the same fact.

2. The route and its scope. Origin point and destination point, stated precisely: a warehouse door in Accra or a port of loading; a warehouse in Lomé or a port of discharge. Door-to-door and port-to-port are different services with different prices, and a quotation that does not say which one it covers cannot be accurate.

3. The commercial frame. For international sales, the agreed Incoterms 2020 rule, which allocates carriage, insurance, and clearance between seller and buyer and therefore determines which legs the quotation must cover.

4. The schedule. When the cargo will be ready, how often it moves, and whether the enquiry concerns one shipment or a recurring programme, because readiness dates and frequency change both the costing and the commercial terms.

5. Special requirements. Temperature control, dangerous goods, out-of-gauge pieces, lifting arrangements, and the documents and permits the movement will need.

The costing professional's first discipline is to refuse to price guesses. Where the enquiry is silent, the professional asks before quoting, because a quotation built on assumed cargo figures

is the first and most common pricing error, and it converts the quoting stage into a dispute in waiting.

KEY CONCEPT

Costing the job: from cost families to cost price

Behind every selling price sits an estimate of what the service will cost the provider. The cost price is the amount incurred in obtaining or delivering a service before adding a selling markup, and building it honestly is the heart of quotation preparation. Lesson 10.1, Components of Total Logistics Cost, supplied the map of cost families; the costing professional now estimates each family for the specific job.

1. Transportation. The line-haul the provider must buy or perform: ocean or air freight with its surcharges, or road transport, priced from the carrier offers actually available, not from memory.

2. Handling. Collection from the customer's premises, consolidation, terminal and warehouse handling, and delivery at destination, each estimated against the cargo's real weight, volume, and handling needs.

3. Documentation and administration. The paperwork the movement requires and the office effort of managing it: bookings, declarations where the scope includes them, permits, and communication across the movement. Lesson 10.1 showed that administration scales with the number of orders and documents, so a recurring programme carries it every month.

4. Time-based risk. Lesson 10.3, Storage, Detention, Demurrage, and Other Charges, showed how waiting becomes invoiced once free time expires. A quotation whose scope touches ports, terminals, or border crossings carries a realistic allowance for that exposure: an estimate of the delay risk the job realistically generates, priced as a provision rather than ignored. Omitting the tail does not remove it; it transfers the tail onto the provider's own account.

5. Own overhead. Lesson 10.1 distinguished direct costs, attributable to one job, from indirect costs shared across the operation. Rent, salaries, systems, and insurance are indirect, and cost allocation spreads a reasoned share of them onto each job. A quotation built from direct costs alone prices the truck and forgets the depot that dispatched it.

The sum of these five estimates is the cost price: the floor below which the provider would be working at a loss.

KEY CONCEPT

Cost price, selling price, markup, and profit margin

Four terms describe the money in every quotation, and professionals keep them exactly apart.

The cost price is the amount incurred in obtaining or delivering a service before adding a selling markup, built in the way the previous concept described.

The selling price is the amount charged to a customer for the product or service: the figure that appears on the quotation.

The markup is the amount added to cost to determine a selling price, commonly expressed as a percentage of cost. A markup of 15 per cent on a cost price of 100 GHS adds 15 GHS and produces a selling price of 115 GHS.

The profit margin is the proportion of selling revenue remaining after specified costs are deducted. In the same example, 15 GHS remains out of 115 GHS of selling revenue, so the margin is 15 divided by 115, approximately 13 per cent.

Markup and margin therefore describe the same money measured on different bases: markup on cost price, margin on selling price. Because the bases differ, the percentages differ, and the same deal can truthfully be called a 15 per cent markup or an approximately 13 per cent margin. Confusing the two is a classic pricing error with real consequences. A manager who promises a 20 per cent margin but calculates it as a 20 per cent markup on cost delivers only about 17 per cent of true margin, and the shortfall repeats on every job priced that way.

No industry rule fixes what the markup should be. Each business sets its own percentage per service, per market, and per risk, as a commercial decision informed by its costs, its competitors, and the value of what it delivers. Every figure used in this lesson is illustrative for teaching purposes; no percentage here is offered as an industry norm.

KEY CONCEPT

Pricing rules: cost-plus and market-based selling prices

With cost price and markup in hand, the selling price follows a stated pricing rule. Two families of rule dominate logistics pricing.

1. Cost-plus pricing. The selling price is the cost price plus a markup calculated on it, so the selling price equals the cost price plus the cost price multiplied by the markup percentage. Cost-plus pricing recovers the provider's costs by construction, which makes it the natural rule wherever the cost build is reliable. Its weakness is inherited directly from that build: if the cost estimate guessed the cargo figures, forgot the overhead share, or ignored the time-based tail, the rule faithfully marks up an error.

2. Market-based pricing. The selling price is anchored to what the market charges and what the service is worth to the buyer, and then tested against the cost price as a floor. Providers read the market differently at different volumes and horizons: a spot rate is a one-off price at the current market, while a contract rate is an agreed rate for a committed volume or period, often indexed to fuel or other moving inputs. Most providers blend the two across their business.

In practice the two rules meet in a discipline: the cost price sets the floor, the market pressures the ceiling, and the markup is the provider's commercial decision in between. An example of the calculation: a project move costs 4,000 GHS in total to deliver, the company's pricing rule adds a

15 per cent markup, so the selling price is 4,000 + (4,000 × 0.15) = 4,600 GHS, quoted subject to the validity and conditions described in the next concepts. All figures are illustrative.

A quotation below cost price is not generosity; it is a forecast loss. The professional checks every selling price against the cost build before the document leaves the office.

KEY CONCEPT

The service scope: inclusions, exclusions, and assumptions

A price without a scope is a number without a meaning. The service scope is the activities and responsibilities included in a proposed service, and stating it precisely is what turns a quotation from a price list into a professional document. Three instruments do the work.

1. Inclusions. The services and charges expressly covered by the quotation: collection from a named point, carriage between named points, handling, the documentation the scope includes, and delivery to a named point. Inclusions answer the buyer's first question: what am I paying for?

2. Exclusions. The services, charges, or responsibilities not covered by the quotation: government duties and taxes, cargo insurance, storage beyond stated allowances, and charges arising after free time expires. Exclusions answer the buyer's sharper question: what is not covered, and who will carry it? An exclusion is not a gap or an evasion; stated in writing, it is information, and it is the buyer's protection against discovering the boundary of the price only after the invoice arrives.

3. Assumptions. The facts on whose truth the price depends: the cargo figures, such as three pallets of 900 kilograms each; the readiness dates; the regularity of a programme; the absence of special handling needs. Assumptions convert uncertainty into stated conditions. If the actual cargo turns out to be six pallets, the quotation has not been cheated; its assumptions no longer hold, and the professional path is a revised quotation, not an argument.

Lesson 10.3 showed how delay charges turn into disputes when nobody established in advance who was responsible for what, and how expensive such disputes are to fight. The quotation is where that establishment happens. This is why professionals describe the quotation as contract-like: it records what is offered, on what scope, at what validity, and under what assumptions, and the discipline of writing assumptions and exclusions down is precisely the discipline that prevents tomorrow's disputes. A professional would rather lose a deal to a clearly stated exclusion than win it on a silence.

KEY CONCEPT

Quotation validity and commercial terms

A quotation prices a service using inputs, carrier rates, fuel, handling tariffs, and exchange rates, that move. Quotation validity is the period during which a commercial offer remains effective under its stated conditions, and it exists because the provider can promise its price only while its

inputs hold. Validity protects both sides. The provider is not bound indefinitely to a price the market has left behind, and the buyer knows exactly how long the offered figures may be relied on. Lesson 5.2 introduced validity as an expiry on the offer; the costing view adds that validity is also a pricing instrument: the provider prices against the market of the validity window, and the buyer must match that window to the reality of its own decision-making. A quotation that expires before the buyer's approval process finishes is worth nothing, however attractive its figures.

Around the validity sit the commercial terms: the conditions governing pricing, payment, service delivery, and related obligations. The recurring elements include the following.

1. Payment terms: when invoices fall due and on what events, such as before carriage, on delivery, or within an agreed number of days after invoice.

2. Currency and rate basis: which currency the quotation uses, and who bears exchange-rate movement within or beyond the validity period.

3. Liability and conditions of service: the terms under which the service is performed, often by reference to the provider's standard trading conditions.

Commercial terms are not decoration. They determine what the quoted figure costs the buyer in practice: a low price payable before carriage, in a currency the buyer must purchase at an uncertain rate, can prove dearer than a higher price on settled terms. Lesson 10.1's warning applies here in commercial form: compare the whole offer, not the headline line.

KEY CONCEPT

The components of a professional freight quotation

The essential components of a professional freight quotation assemble the concepts so far into one document. A complete quotation contains the following.

1. Parties and reference. Who is quoting, to whom, the enquiry it answers, and the date of issue, which starts the validity clock.

2. Scope statement. The route and service level in plain terms: named origin point, named destination point, door-to-door or port-to-port, and the Incoterms 2020 basis where an international sale is involved.

3. Itemised charges. A line for each cost family: carriage, handling, documentation, and the other charges the scope includes. Itemisation mirrors the provider's cost build and allows the buyer to compare offers line by line; a single opaque total forces the buyer to evaluate a mystery.

4. Total and currency. The sum of the lines, in a stated currency.

5. Validity period. The date until which the offer stands, stated explicitly.

6. Payment terms. When and how payment falls due.

7. Inclusions. The services and charges covered, summarised so the scope can be read at a glance.

8. Exclusions. The charges and responsibilities not covered, listed explicitly.

9. Assumptions. The cargo figures, dates, and conditions on which the price depends.

10. Conditions. The commercial terms and trading conditions that govern performance.

The list is deliberately unglamorous, and its value is that every later disagreement has a written home. If the cargo figures change, the assumptions paragraph governs. If free time runs out, the exclusions paragraph governs. If rates move, the validity paragraph governs. The worked examples of this lesson build a quotation with exactly these components for a cross-border programme, and then apply a buyer's checklist to two quotations that differ precisely in how many of these components they contain. Components are not paperwork around the price; they are the price made evaluable.

KEY CONCEPT

Evaluating a quotation: normalise the scope, check the clock, test completeness

Quotation evaluation is the assessment of a commercial offer against cost, service, risk, and operational requirements. Lesson 5.2 taught the beginning of the skill, reading beyond a headline price; the costing professional extends it into a checklist that can be applied to any offer, line by line. Rate comparison, the evaluation of competing prices and commercial conditions, begins only after three preparatory steps.

1. Normalise the scope. Offers are comparable only when they describe the same service. A door-to-door quotation and a port-to-port quotation differ by whole legs: collection at origin, terminal and documentation charges at both ends, and delivery at destination, the very lines Lesson 7.4, Shipping Charges, Freight Rates, and Surcharges, took apart. Before comparing figures, restate each offer on the same scope: what must be added to the narrow offer to equal the wide one, and what the wide one already contains. Only equalised totals are prices for the same thing.

2. Check validity against the decision timeline. Note each offer's validity and set it against how long the buying decision will actually take. A quotation that lapses before the order is placed is worth nothing, whatever its headline; and an offer valid for ten days quietly demands a decision the buyer may not be able to make.

3. Test completeness. Read every offer for what it does not say. Is insurance included or excluded? Destination terminal handling? Customs fees and agency charges? Storage if the consignee is slow to collect? Each silence is one of two things: an exclusion waiting to be confirmed, or a hidden cost that will surface as an invoice after booking. The professional

converts every silence into a written question before comparing prices, because hidden-cost risk hides exactly where the document is quiet. The habit Lesson 10.2, Freight Costing and Landed-Cost Calculations, enforced on landed-cost builds, that a build which omits lines flatters the decision, applies equally to a quotation under evaluation.

Only when two offers have been normalised to one scope, confirmed live for the decision window, and completed to the same visible content do their numbers describe the same purchase.

KEY CONCEPT

Beyond the price lines: the offerer, sensitivity, and the total-cost view

Three further steps complete the checklist.

4. Assess the offerer. A quotation is also a sample of its provider: whether the document is precise or vague, whether assumptions are stated or smuggled, whether the provider asked questions before pricing. Beyond the paper, the professional weighs capability and reliability: can this provider perform the stated scope, with the equipment, network, agency arrangements, and references to show it? For a recurring service, the instrument is the service-level agreement, an agreement defining specified service expectations, performance measures, and responsibilities. Its typical components, scope of services, key performance indicators with targets, measurement and reporting cadence, remedies or service credits, governance and review, and exit terms, convert promises into measurable commitments. One line forward: the full discipline of selecting and contracting a provider belongs to Module 15, Strategic Logistics Management; at the quotation stage the professional's job is to weigh the evidence the offer itself provides.

5. Test sensitivity. Ask of every line what happens if reality moves. If volumes fall below the assumptions, does the price survive? If readiness slips, which time-based charges from Lesson 10.3 come alive, and on whose account? If the market moves, spot lines reprice at renewal while contract lines hold to their terms, which is why the spot-versus-contract structure of an offer is itself an allocation of risk between the parties.

6. Apply the total-cost view. Lesson 10.1's lens closes the checklist: the cheapest quotation is an input, not a conclusion. Transit time, reliability, service coverage, and commercial risk are part of what the money buys, and this lesson's criteria, total cost, reliability, transit time, service coverage, and commercial risk, are weighed together rather than one at a time.

The checklist also exposes the recurring pricing errors made on the provider's side of the desk: quoting on guessed cargo figures; omitting the overhead share; confusing markup with margin; pricing from a rate base that has since expired; leaving the time-based tail unpriced; and trusting an unstated exclusion to remain unstated. Every one of them is cheaper to fix before the quotation is issued than after it is accepted.

KEY CONCEPT

The negotiation moment: clarify, negotiate openly, document

Between receiving a quotation and accepting one lies the negotiation moment, and the professional handles it in three motions.

First, clarify before you bargain. Questions about scope, assumptions, and completeness are not negotiation; they establish what is actually being priced, and providers expect them. Ask before you bargain: a discount negotiated on a misunderstood scope is an agreement to disagree. The buyer who has normalised the scope and tested completeness, the checklist's first three steps, knows exactly which silences need answering and which differences between offers are real.

Second, negotiate openly and reprice scope, not silence. Where the buyer genuinely needs something the quotation excludes, insurance, storage flexibility, a wider cargo allowance, the professional path is to ask for it and accept a revised price for a revised scope. Unpriced assumptions are tomorrow's invoices: anything the buyer persuades itself is probably included will arrive later as a charge nobody budgeted, with none of the negotiating leverage that an open request would have had. The same discipline binds the provider. Absorb an unstated assumption quietly and the cost is real; requote and the cost becomes a number the customer can accept or refuse.

Third, document the accepted scope. The purchase order or booking that follows acceptance should mirror the quotation: same scope, same figures, same assumptions, same validity terms. When the documents mirror each other, later invoices can be checked against an agreed standard mechanically, and the procurement habit of matching order, service received, and invoice before payment becomes routine rather than forensic. Where circumstances change after acceptance, the change is handled in writing, as a revised quotation or a written amendment, never as a conversation both parties later remember differently. In logistics pricing, the memory that counts is the one on paper.

WORKED EXAMPLE

Building the quotation: a monthly Accra to Lomé consolidation programme

A manufacturer of packaged food products in Accra ships a regular monthly consignment to its distributor in Lomé, Togo, and asks a freight forwarder in Accra to quote for the programme: collection from the factory, consolidated road carriage to Lomé on the coastal corridor that links Ghana with Togo, and delivery to the distributor's warehouse. All figures in this example are illustrative for teaching purposes.

Step 1: Fix the facts of the enquiry. Three pallets per month, 900 kilograms each, standard stackable pallets, non-dangerous goods, collected in one pick-up each month in an agreed week, delivered door to door, recurring for the life of the offer. The costing officer confirms each point with the client before pricing: the entire cost build hangs on them.

Step 2: Cost each family for one month. Transport, the allocated share of the consolidated line-haul: 2,700 GHS, that is 900 GHS per pallet. Handling, collection, consolidation handling, and delivery in Lomé: 900 GHS. Documentation for the cross-border movement: 600 GHS. Time-based risk, an allowance for border and terminal waiting on a road programme: 300 GHS. Overhead allocation, a share of the depot, salaries, and systems: 600 GHS. Total cost price: 5,100 GHS per month, or 1,700 GHS per pallet.

Step 3: Apply the pricing rule. The company's rule for this lane is a 15 per cent markup on cost: 5,100 × 0.15 = 765 GHS, giving a selling price of 5,865 GHS per month, or 1,955 GHS per pallet. This is a 15 per cent markup but only about a 13 per cent margin: 765 of every 5,865 GHS invoiced.

Step 4: Write the commercial layer. Validity: 30 days from the date of issue. Payment terms: invoiced monthly, payable within 14 days. Inclusions: collection in Accra, consolidation, road carriage, cross-border documentation, and delivery in Lomé. Exclusions: import duties and taxes in Togo and import clearance at destination, which remain the consignee's responsibility; cargo insurance, quoted separately on request; storage at destination beyond two working days. Assumptions: three pallets per month at up to 900 kilograms each, stackable and non-dangerous, ready in the agreed collection weeks; quantities outside these figures will be requoted.

Step 5: Assemble the document, stating the parties and date of issue, the door-to-door scope, the monthly charge of 5,865 GHS, the validity, the payment terms, and the inclusions, exclusions, and assumptions, each in its own paragraph.

What does this example demonstrate?

Every figure on the finished document traces back to a costing decision, and every condition protects the offer from a future argument. Key lesson: a professional quotation is a cost build plus a commercial layer, and the commercial layer is what makes the document an offer rather than a number.

WORKED EXAMPLE

Evaluating two quotations for one project move, Mumbai

A manufacturer in Mumbai must import a production line, twelve crates totalling 18 tonnes, from a supplier near Shanghai, discharged at Mumbai and delivered to the factory. Two freight forwarders return quotations. All figures in this example are illustrative for teaching purposes.

Forwarder A's quotation is itemised. Ocean freight including surcharges: 2,600 US dollars. Origin handling and documentation: 350 US dollars. Destination terminal handling: 450 US dollars. Destination documentation and agency fee: 300 US dollars. Inland delivery, port to factory: 500 US dollars. Total: 4,200 US dollars. Validity: 30 days. Exclusions, stated: customs duty and taxes, cargo insurance, and any detention or demurrage beyond the free time allowed. Assumptions, stated: twelve crates, 18 tonnes, ready for collection in week six.

Forwarder B's quotation is one line: door-to-door delivery, 3,900 US dollars, all-inclusive. Validity: 10 days. No exclusions, no assumptions, no cargo basis.

The buyer applies the checklist.

Step 1: Normalise the scope. A's total covers the port-to-port movement plus named destination lines; B claims door-to-door but does not say whether destination terminal handling and destination documentation sit inside its total. Until B answers, the two prices are not prices for the same service.

Step 2: Check the clocks. The buyer's capital approval takes about three weeks. B's offer expires in ten days, before the decision can be made; A's thirty days cover it.

Step 3: Test completeness. A names duty, insurance, and free-time excess as exclusions; B is silent, and silence is not inclusion. If B's unstated destination lines merely cost the same as A's comparable lines, an illustrative supposition, B's true comparable total is 3,900 plus 750 US dollars, about 4,650, some 450 US dollars above A, and the buyer discovers this only after booking.

Step 4: Assess the offerers. A asked for the cargo list, stated its assumptions, and attached references from similar projects; B returned a price and nothing else.

Step 5: Test sensitivity. If the crates exceed 18 tonnes, B's price has no stated basis on which to be repriced, while A's assumptions make the position testable.

Step 6: Weigh total cost. A's 4,200 US dollars is transparent and live for the decision window; B's 3,900 is a headline attached to an unevaluable offer.

What does this example demonstrate?

The vague quotation need not be dishonest, but it is unevaluable, and an unevaluable offer transfers all of its risk to the buyer. Key lesson: apply the checklist before the prices; prices become comparable only when the offers describe the same service on the same stated conditions.

WORKED EXAMPLE

The validity miss: six weeks against a 14-day offer

An importer in Accra buys general merchandise overseas and asks its freight forwarder for a quotation for the next consolidated shipment. The forwarder prices the job on the rates then available and issues a quotation valid for 14 days. All figures in this example are illustrative for teaching purposes.

The quotation reaches the importer's managing director just before he travels. Approval waits for his return, then for the finance meeting, then for the overseas supplier's own confirmation. Six weeks after issue, the importer writes: we accept your quotation, please proceed.

The forwarder now holds an acceptance of an offer that expired more than four weeks earlier. In the meantime the market has moved: the shipping line serving the lane has announced a general rate increase, an across-the-board upward adjustment of base freight rates, and the forwarder's own input costs now exceed the quotation. The quoted price no longer describes a service the forwarder can buy at a profit.

Who bears the movement? The quotation's own terms answer the question: the offer stood for 14 days and then lapsed. The importer cannot insist on a price whose validity it allowed to pass, and the forwarder is not obliged to perform at a stated loss, although honouring a lapsed quotation as a gesture to a valued customer remains the forwarder's commercial choice to make, not its obligation.

How does the professional handle it? The forwarder responds promptly and in writing: it explains that the offer has expired, shows the movement in its inputs, and issues a fresh quotation at the current market. The importer now has real choices: accept the new figures, exercise rate comparison and seek alternative offers, or discuss a contract-rate arrangement, an agreed rate for a committed volume over a period, which would buy stability for a recurring programme at the price of a commitment. What the parties do not do is what they would most regret: the forwarder silently honouring a loss it cannot afford, or silently repricing the old document without saying so.

What does this example demonstrate?

Both parties could have prevented the moment. The importer could have matched its approval timeline to the validity, or asked for an extension before expiry; the forwarder could have flagged that rates on the lane were due to move. Key lesson: validity periods protect both sides, and the professional honours them or renegotiates openly, in that order.

IMPORTANT NOTE

A quotation is an offer, not a contract of carriage

A misconception follows quotations out of the classroom into real trading: that accepting a quotation is the same thing as having a contract of carriage, or that a quotation binds the provider until the customer releases it. Neither is true, and the difference matters operationally.

A logistics quotation is a commercial offer: a statement of proposed charges and conditions for specified services. It becomes binding through acceptance on its terms, and its terms include its validity period and its assumptions. Accept the offer within validity, on the stated basis, and the parties have a commercial agreement for that scope at that price. Accept it after validity, or on a different basis, and there is nothing yet to bind: the offer has lapsed, and the acceptance is really a new negotiation.

The contract of carriage is a separate instrument. It arises from the booking and is evidenced by the transport document, which Module 5 treated in detail; the quotation is neither of these. That is also why a quotation can be honest, complete, and professional, and still cease to be

available: its inputs moved, its window closed, and the offer, unlike a carriage already performed, simply ended.

The practical rules follow. Providers honour their validity periods or renegotiate openly before them. Buyers accept within validity or ask for a fresh offer. Neither side treats a lapsed quotation as a grievance. The document that binds is the one both parties accepted on the day.

IMPORTANT NOTE

Never quote or accept on unstated assumptions

The sentence worth carrying out of this lesson is short: unpriced assumptions are tomorrow's invoices. It applies on both sides of the quotation.

The provider who quotes on guessed cargo figures, an assumed readiness date, or an unexamined belief that no special requirements apply has not avoided the risk; the provider has bought it. The goods arrive heavier than assumed, the collection slips into a rate change, the cargo needs lifting equipment nobody priced, and each discovery arrives as a choice between absorbing a loss and starting a dispute with a new customer. Every one of these outcomes was visible at the enquiry stage, which is why the first discipline of quotation preparation is to ask before pricing and to write the answers down as assumptions.

The buyer runs the same risk in reverse. Accepting a quotation whose assumptions are hidden, whose exclusions are silent, and whose validity does not match the decision timeline means accepting obligations that no one has priced. Lesson 10.3's disputes were born in silences of exactly this kind; the quotation is where they are strangled, by an inclusions list, an exclusions list, and an assumptions list that the buyer has read and tested rather than merely filed.

Scope changes after acceptance are normal and legitimate. Cargo grows, dates move, programmes expand. The professional response is never silence in either direction: it is a revised quotation or a written amendment, so that the accepted scope and the invoiced scope remain the same document. In logistics pricing, what is written down is what was agreed; everything else is a forecast of an argument.

Practice questions (Lesson Checks)

Practice Question 1

A costing officer at a freight forwarder in Accra builds a cost price of 8,000 GHS for a project move. The company's pricing rule adds a 25 per cent markup on cost to determine the selling price. All figures are illustrative for teaching purposes.

Which option best describes the figures that should appear in the quotation?

1. A selling price of 10,000 GHS, which the quotation may describe as earning a profit margin of 25 per cent of the selling price.

2. A selling price of 10,000 GHS, which earns a profit margin of 20 per cent of the selling price, because markup is calculated on cost price while margin is measured against the selling price. ✓

3. A selling price of approximately 10,667 GHS, because a 25 per cent requirement must always be applied by dividing the cost price by 0.75.

4. A selling price of 8,000 GHS, because the markup is the provider's internal profit and is never included in the price shown to the customer.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The markup rule works on the cost price: 25 per cent of 8,000 GHS is 2,000 GHS, so the selling price is 10,000 GHS. The profit margin is then measured on the selling price: 2,000 GHS remaining out of 10,000 GHS invoiced is 20 per cent. Markup and margin describe the same money on different bases, so the two percentages differ, and a professional document keeps them distinct.

Answer 1 restates the classic confusion this lesson dismantles: a 25 per cent markup on cost is a 20 per cent margin on selling price, not a 25 per cent one. Answer 3 applies a different pricing rule; dividing the cost price by 0.75 is how a 25 per cent margin on the selling price would be built, which is not the company's markup rule. Answer 4 denies the purpose of the markup: the selling price exists to recover the cost and earn the margin, so it must include the markup.

The takeaway: markup lives on the cost price, margin lives on the selling price, and the two figures are both true only when each is measured on its own base.

Practice Question 2

A buyer in Kumasi receives two quotations for the same import shipment. The first quotes 5,200 US dollars, fully itemised, with its exclusions listed and a validity of 30 days. The second quotes 4,750 US dollars as an all-in price, lists no exclusions, and is valid for 10 days. The buyer's internal approval will take about four weeks. All figures are illustrative for teaching purposes.

Which option best explains the professional conclusion?

1. The all-in quotation is the better commercial choice, because a single inclusive price transfers every risk of the movement to the provider.

2. The all-in quotation cannot yet be compared with the itemised one, because its scope is unverified, its silences may conceal charges, and its validity expires before the buyer's decision can be made. ✓

3. The two quotations are equally valuable, because the buyer can accept the all-in price whenever the internal approval finishes, regardless of its stated validity.

4. The itemised quotation is automatically the more reliable offer, because any provider who itemises charges always operates better than one who does not.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The checklist of this lesson is applied before the prices are compared. The all-in offer has not stated what its price covers, so the places where destination charges, insurance, or customs fees would sit remain untested; an unstated scope may hide costs that surface as invoices after booking. Its validity of 10 days also expires long before the buyer's four-week approval concludes, so even its headline figure may no longer be available when the decision is made. Until the scope is normalised and the validity is extended, the lower number describes an unevaluable offer.

Answer 1 fails because silence transfers risk to the buyer rather than away from them: an all-in price with no exclusions stated means nobody knows what happens when an unlisted charge appears. Answer 3 ignores quotation validity: an offer lapses at the end of its stated period, and accepting afterwards begins a new negotiation instead of conferring a right. Answer 4 over-generalises; itemisation is encouraging evidence of transparency in this scenario, but the checklist weighs scope, validity, completeness, the offerer, sensitivity, and total cost together rather than awarding the decision to a document style.

The takeaway: normalise the scope, check the clock, and test completeness before letting headline prices compete.

Practice Question 3

An exporter in Kumasi receives a freight quotation for six pallets of shea butter shipped to a customer overseas, valid for 14 days. Its board approves the shipment in week six and writes to accept the original figures. In the meantime, the costs underlying the quotation have risen, and the forwarder can no longer deliver the quoted scope at the quoted price.

Which statement best describes the position?

1. The forwarder must honour the quotation, because a quotation remains binding on the provider until the customer decides to release it.

2. The exporter owes a late-acceptance penalty to the forwarder, calculated from the day the quotation expired.

3. The offer lapsed at the end of its validity, so the exporter cannot insist on the original price; the professional path is a fresh quotation and open renegotiation, which protects both sides. ✓

4. A telephone call accepting the quotation during the first week would have fixed the price permanently, so the exporter should have accepted by phone and organised the shipment later.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Quotation validity is the period during which the offer remains effective under its stated conditions, and it protects both sides: the provider is not bound indefinitely to a price its market has left, and the buyer knows exactly how long the figures may be relied on. When the exporter allowed the 14 days to pass, the offer ended, and a later acceptance cannot revive it. The forwarder may choose to honour the old figures as a commercial gesture, but that is a choice, not an obligation; the professional handling is to say so promptly, explain the movement in inputs, and issue a fresh quotation, leaving the exporter free to accept it, compare alternative offers, or discuss a contract-rate arrangement for stability.

Answer 1 reverses the nature of the document: a quotation is an offer with its own terms, not an open-ended commitment that only the customer can end. Answer 2 invents a rule that does not exist; the consequence of a lapse is renegotiation, not a penalty. Answer 4 misreads validity in the opposite direction: an acceptance made within validity creates an agreement on the quoted terms for that service, not a permanent price fix against future market movement.

The takeaway: professionals match validity to their decision timelines, and where an offer lapses, the remedy is honest repricing, not grievance.

Fact-check appendix (audit notes — not for entry)

Definitions of logistics quotation, cost price, selling price, markup, profit margin, service scope, quotation validity, inclusions, exclusions, rate comparison, service-level agreement, commercial terms, and quotation evaluation follow the lesson brief's key terms, defined at first use and used in the body. — source: lesson brief 10.4, Key Terms (curriculum database).

A quotation is a commercial offer that becomes binding through acceptance on its terms; the contract of carriage arises separately from the booking and is evidenced by the transport document. — source: general commercial and carriage-law principles (offer and acceptance), consistent with research sheet trade-transport-standards.md section 1 note that trade terms do not themselves constitute a carriage contract.

Incoterms 2020 allocate transport tasks, costs, and risk between buyer and seller, determining who arranges carriage, insurance, and clearance; they do not transfer ownership or constitute a carriage contract. — source: research sheet trade-transport-standards.md section 1; research sheet logistics-scm-frameworks.md section 10.

Freight quotation line composition concept: base ocean freight plus surcharges, terminal handling at both ends, documentation, and security-type charges (names only, no amounts). — source: research sheet trade-transport-standards.md section 7 (safe phrasings).

A general rate increase is an across-the-board upward adjustment of base freight rates, typically announced for contract seasons or capacity management; taught as a concept, no rates or dates. — source: research sheet trade-transport-standards.md section 7.

Free time precedes time-based charges; demurrage, detention, and storage are purely contractual tariffs with no standard international amounts; this lesson teaches the concept and prices only illustrative provisions. — source: research sheet trade-transport-standards.md section 11.

Spot rate is a one-off price at the current market; contract rate is an agreed rate for a committed volume or period, often indexed; providers blend both. — source: research sheet logistics-scm-frameworks.md section 10.

Typical SLA components: scope of services; KPIs with targets; measurement and reporting cadence; remedies or service credits; governance and review; exit terms. — source: research sheet logistics-scm-frameworks.md section 10 (taught as components, not a fixed canon).

Provider selection weighs service scope fit, coverage, track record and references, cost model, and related criteria; full treatment deferred to Module 15 (Strategic Logistics Management, whose lessons include logistics strategy and service-provider selection, and contract management and service-level agreements). — source: research sheet logistics-scm-frameworks.md section 10; module 15 lesson briefs.

Chargeable weight is priced on the greater of actual weight and volumetric weight (referenced by name only, no re-teaching of the arithmetic). — source: research sheet trade-transport-standards.md section 8; lesson brief 5.2.

Matching order, service receipt, and invoice before payment referenced as standard procurement control (three-way match). — source: research sheet logistics-scm-frameworks.md section 3.

Cost families, direct and indirect costs, cost allocation, the hidden-cost tail, and the total-cost lens are Lesson 10.1 material, referenced by name only. — source: lesson draft 10.1 (Components of Total Logistics Cost); research sheet logistics-scm-frameworks.md sections 6 and 7.

Time-based charges, free time, expected-charges budgeting, and dispute handling are Lesson 10.3 material, referenced by name only. — source: lesson draft 10.3 (Storage, Detention, Demurrage, and Other Charges); research sheet trade-transport-standards.md section 11.

Landed-cost builds omit lines at the decision's peril: Lesson 10.2 referenced by name only. — source: lesson draft 10.2 (Freight Costing and Landed-Cost Calculations).

A coastal road corridor links Ghana with Togo to the east (context for the Accra to Lomé example); no border-post names stated. — source: research sheet ghana-trade-customs.md section 8 (safe phrasing).

Markup-versus-margin arithmetic (15 per cent markup is about a 13 per cent margin; 20 per cent markup is about a 17 per cent margin; 8,000 GHS plus a 25 per cent markup is 10,000 GHS, a 20 per cent margin) is arithmetic, not an industry claim; no industry-standard margin or markup percentage is asserted anywhere. — source: WRITING_SPEC.md fact-safety rule 5.1; arithmetic verified.

All companies, quantities, prices, markups, margins, validity periods, and payment terms in the worked examples and MCQs are illustrative for teaching purposes and are labelled as such; no market rate levels, real tariffs, or benchmark percentages are claimed, and no real company is attached to factual claims. — source: WRITING_SPEC.md rules 5.1 and section 4; the research sheets' DO NOT CLAIM lists.

References to Lesson 5.2 (Freight Quotations and Shipment Booking), Lesson 7.4 (Shipping Charges, Freight Rates, and Surcharges), Lesson 10.1 (Components of Total Logistics Cost), Lesson 10.2 (Freight Costing and Landed-Cost Calculations), Lesson 10.3 (Storage, Detention, Demurrage, and Other Charges), Module 5 transport documents, and Module 15 (Strategic Logistics Management) are by name only, with no re-teaching. — source: WRITING_SPEC.md section 7; lesson briefs and drafts cited.

Module 10 Assessment — 12 questions

Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.

Question 1 (tests lesson 10.1)

1

A distributor of imported ceramic tableware in Takoradi buys its packaging as sets, one carton with inner crating per set, from Supplier P at 18 GHS a set. A rival supplier, Q, offers equivalent-looking sets at 14 GHS. The distributor ships 1,000 cartons a month and switches to Q. Three months later, the share of cartons arriving damaged has risen from about one in a hundred to about four in a hundred, and each damaged carton generates about 350 GHS of replacement transport, re-handling, credit notes and claims administration. All figures are illustrative for teaching purposes.

Which option best evaluates the switch?

1. The switch raised total logistics cost: packaging is a damage-prevention cost, and the switch's tail, roughly thirty extra damaged cartons a month at about 350 GHS each, some 10,500 GHS, outweighs the 4,000 GHS monthly saving by about 6,500 GHS. ✓

2. The switch lowered total logistics cost, because returns, re-shipments and claims are estimates rather than invoiced amounts, and only invoiced amounts enter the total.

3. The two effects cannot be compared, because packaging is a direct cost of each consignment while damage losses are indirect costs of the operation as a whole.

4. The switch is justified, because the per-carton packaging price fell and unit cost is the measure that matters; breakage is a customer-service matter outside the costing decision.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

The lesson teaches that packaging expenditure is not only a materials cost but a damage-prevention cost, and that skimping on it usually shows up later in the failure families. The arithmetic runs through the whole lens. The switch saved 4 GHS a set across 1,000 sets a month,

some 4,000 GHS. The damage rate rose by about three cartons in every hundred, some thirty extra damaged cartons a month, each carrying about 350 GHS of replacement transport, re-handling, credit notes and claims administration, some 10,500 GHS a month. The visible saving of 4,000 is outweighed by the invisible tail of 10,500, so the switch makes the operation dearer by about 6,500 GHS a month, before counting customers who quietly reduce their orders after repeated breakages.

Answer 2 applies the misconception the module exists to dismantle: failure costs are real money whether or not a single invoice names them, scattered as they are across transport, wages and write-offs. Answer 3 uses classification as an evasion; the direct and indirect labels describe attribution, not comparability, and total-cost thinking exists precisely to compare effects across families. Answer 4 optimises the single visible line and waves the tail away as a service matter; the fallacy of optimising one line is exactly what the cheaper-looking per-carton price conceals.

The takeaway: before signing for the cheaper input, price the tail the choice makes likely and compare the whole effect on the total, not the movement of one line.

Question 2 (tests lesson 10.1)

2

A distributor of consumer electronics in Accra normally buys 500 units a month from its overseas supplier at 1,000 GHS a unit. The supplier offers a five per cent price reduction if the distributor takes 3,000 units in one six-month bulk purchase at 950 GHS a unit, paid upfront. The purchasing manager presents the 150,000 GHS discount as pure saving. The finance officer estimates that carrying the much larger stock for the longer period costs about 190,000 GHS in capital tied up, insurance, obsolescence as newer models arrive, and shrinkage. All figures are illustrative for teaching purposes.

Which statement best evaluates the offer?

1. The offer is free money: the goods are simply bought earlier, the company's warehouse already exists, and no invoice charges the company for owning its own stock.

2. The offer raises total cost: the 150,000 GHS discount is real, but the holding side is real too, and at about 190,000 GHS it outweighs the discount by some 40,000 GHS, even though no invoice will ever name that figure. ✓

3. Holding cost applies only while goods wait in a public warehouse under a storage tariff; stock in the company's own building costs nothing beyond the rent already paid.

4. The only holding cost relevant here is the warehouse rent, which does not change, so the discount stands at the full 150,000 GHS.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Order size is a trade-off, not a free win: the third classic trade-off of Lesson 10.1 balances ordering savings against carrying cost. The discount side is real, 50 GHS less on each of 3,000 units, some 150,000 GHS. The carrying side is assembled from equally real components: the additional capital tied up in slow-moving stock that cannot fund anything else, insurance on goods sitting for months, obsolescence as newer models arrive, and shrinkage. The finance officer prices that side at about 190,000 GHS, so the deal is about 40,000 GHS dearer in total. Nothing about that conclusion requires an invoice addressed to the stock: inventory holding cost is the standing proof that money can be fully spent without a document naming it.

Answer 1 is the argument Lesson 10.1's own knowledge check dismantles: goods already paid for are capital already committed, which is the cost, not the absence of one. Answer 3 confuses the inventory holding family with the facility tariff charges treated in Lesson 10.3; holding cost accrues in the company's own building from the first day. Answer 4 mistakes the boundary between families: an unchanged rent means the warehousing family is unchanged, while the holding family, driven by value and time held, has grown with every extra unit on the shelf.

The takeaway: compare a purchasing saving against the whole holding tail it creates, and treat the assembled holding cost as a first-class figure in the decision.

Question 3 (tests lesson 10.1)

3

A finance officer at a distribution company in Tamale must classify four warehouse costs: casual discharge labour hired by the tonne as each consignment arrives; special crating built for one out-of-gauge machine; monthly lease instalments on the forklift fleet; and the annual licence for the warehouse's management software.

Which option best classifies them?

1. All four are indirect costs, because each of them supports the warehouse operation as a whole rather than any single consignment.

2. The forklift lease and the software licence are direct costs because invoices arrive for them every month, while the discharge labour and the crating are fixed costs because they arise only occasionally.

3. The discharge labour and the special crating are direct costs, the labour varying with the tonnage handled, while the forklift lease and the software licence are indirect costs that remain largely unchanged within the company's current operating range. ✓

4. All four are variable costs, because each of them could be reduced or cancelled if the company's activity fell.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The two sortings ask different questions. Direct versus indirect asks whether the expense attaches to one specific shipment, service or activity: the discharge labour is hired for named consignments as they arrive and the crating is built for one machine, so both are direct; the forklift fleet and the management software serve every pallet in the building and can only be shared across the operation through cost allocation, so both are indirect. Fixed versus variable asks whether the expense moves with the level of activity: labour hired by the tonne rises and falls with throughput, while lease instalments and an annual licence remain relatively unchanged within the relevant operating range. The crating sits with the labour on the direct side even though it arises only occasionally, because occasional is not the definition of fixed; the fixed test is about staying unchanged across a period and range.

Answer 1 fails because supporting the whole operation is exactly what makes a cost indirect, and the labour and crating do not have that character. Answer 2 repeats the visibility error: an invoice arriving monthly says nothing about attributability, and frequency of invoicing is not the fixed-versus-variable test. Answer 4 stretches variable to mean cancellable; the test is whether the amount changes with activity, and the lease and licence amounts do not, within range.

The takeaway: ask of every cost both questions, attributable to what, and driven by what volume, and the two answers together place it correctly.

Question 4 (tests lesson 10.2)

4

A forwarder in Accra tenders an air consignment of five identical cartons of machine components. Each carton measures 120 by 100 by 80 centimetres and weighs 96 kilograms gross. Volumetric weight is computed at the IATA standard conversion factor of 6,000 cubic centimetres per kilogram, as taught in this module. The airline's tariff quotes an illustrative GHS 90 per chargeable kilogram for the lane. All figures are illustrative for teaching purposes.

Which option states the consignment's chargeable weight and the resulting freight charge?

1. 480 kg and GHS 43,200, because air freight is charged on the gross weight that the airline's scales record.

2. 640 kg and GHS 57,600, because the chargeable weight per carton is the average of its gross weight and its volumetric weight.

3. 160 kg and GHS 14,400, because the five identical cartons are charged as a single piece on one volumetric weight.

4. 800 kg and GHS 72,000, because each carton's volumetric weight of 160 kilograms exceeds its gross weight of 96 kilograms, so every carton is charged on 160 chargeable kilograms. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Chargeable weight in air freight is the higher of the actual gross weight and the volumetric weight, computed piece by piece under the applicable factor. Each carton measures 120 by 100 by 80 centimetres, a volume of 960,000 cubic centimetres, which divides by 6,000 to a volumetric weight of 160 kilograms. That exceeds the 96-kilogram gross weight, so each carton is charged on 160 kilograms and the five-carton consignment on 5 x 160 = 800 chargeable kilograms. At the illustrative GHS 90 per chargeable kilogram, the freight is 800 x 90 = GHS 72,000. The pricing basis, not the scales alone, decides the number.

Answer 1 applies the dense-cargo habit to bulky cargo: gross weight governs only where it is the greater of the two, and here it is not. Answer 2 invents an averaging step that exists in no tariff; the rule takes the greater figure precisely because the airline is paid for whichever scarce resource the cargo consumes, here hold space rather than payload. Answer 3 forgets that the quotation prices a consignment: five identical pieces each carry their own chargeable weight, and charging one piece's volume against the whole shipment understates the freight by four-fifths.

The takeaway: compute volumetric weight per piece, compare it with the gross weight per piece, charge on the greater of the two for every piece, and check the conversion factor the applicable tariff actually specifies.

Question 5 (tests lesson 10.2)

5

A distributor in Cape Coast imports 400 cartons of packaged toiletries. The costing sheet lists, all figures illustrative for teaching purposes: goods value GHS 96,000; ocean freight GHS 8,400; cargo insurance GHS 1,600; import duty, shown as an illustrative assessment placeholder, GHS 9,200; combined taxes and levies, illustrative placeholders, GHS 19,600; terminal handling and port charges GHS 3,900; inland transport to the warehouse GHS 4,500; clearance and agency fees GHS 2,800; and a storage and contingency allowance of GHS 2,000. A colleague proposes deleting the allowance as only an estimate, and adding a profit line of GHS 4,000, arguing that the goods must also carry their profit.

Which option correctly states the landed cost and the evaluation of the colleague's proposal?

1. GHS 146,000, or GHS 365.00 per carton, because a contingency allowance is a forecast rather than an expenditure and must wait until an invoice exists.

2. GHS 152,000, or GHS 380.00 per carton, because the profit line makes the figure honest about what the business needs to earn from the consignment.

3. GHS 148,000, or GHS 370.00 per carton; every line the buyer will really pay belongs inside the build, including a stated tail allowance, while profit is added on top when the selling price is set, never inside the cost it is meant to recover. ✓

4. GHS 150,000, or GHS 375.00 per carton, because the colleague is right on both counts: estimates never belong in a landed cost, and profit is part of what the goods cost.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The build is added in order: 96,000 plus 8,400 plus 1,600 plus 9,200 plus 19,600 plus 3,900 plus 4,500 plus 2,800 plus 2,000 gives GHS 148,000, and dividing by 400 cartons gives the unit landed cost of GHS 370.00. The colleague's two challenges fail in opposite directions. The storage and contingency allowance belongs inside: the discipline of Lesson 10.2 is that a landed-cost model is only as honest as its tail, and an allowance for the unknown, stated next to the total, is information rather than error; deleting it forecasts a best case and presents it as a measurement. The profit line belongs outside: landed cost answers what the goods cost to place ready for sale, and profit is added on top when the selling price is set. Building profit into the very cost the price must recover destroys the figure's meaning as the floor beneath pricing.

Answers 1 and 4 both act on the deletion of the allowance and so understate the build. Answers 2 and 4 both admit the profit line and so convert a cost measure into an aspiration. Note also what the landed cost is not: the customs value on which the duty was computed is a legally determined base, and the landed total carries destination and tail lines the assessment never shows.

The takeaway: every line the buyer will really pay belongs in the build, the stated tail included, and nothing that is not a cost does.

Question 6 (tests lesson 10.2)

6

An importer in Accra buys machinery spare parts valued at 20,000 US dollars on FOB terms and builds the landed cost at an assumed illustrative rate of 12 cedis to the dollar, pricing the goods

line at GHS 240,000 and setting selling prices from the finished build. The ocean freight and a fuel surcharge are also quoted in US dollars. The parts arrive and are paid for three months later, when the assumed illustrative rate is 13 cedis to the dollar. All figures are illustrative for teaching purposes.

Which statement best describes what the currency movement has done to the landed cost, and the professional response?

1. Every dollar line converts at the rate ruling when payment is made: at 13 rather than 12 cedis to the dollar the goods line alone costs some 20,000 GHS more and the dollar-quoted freight rises proportionally, while locally quoted cedi lines do not move, so the build, which is a snapshot dated to its rate, must be re-run before the selling prices are relied on. ✓

2. Nothing has changed, because the invoice still reads 20,000 US dollars and the landed cost was fixed on the day the order was placed.

3. Only the duty and tax lines can change, because government charges respond to currency movement while commercial prices are fixed by contract.

4. The importer should now price from the customs value instead of the landed cost, because currency movement makes landed-cost builds unreliable.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A landed-cost build is a snapshot dated to the rate it used. Every foreign-currency line converts at the rate ruling when the money is actually paid, not when the order was placed. At 13 cedis to the dollar instead of 12, the goods line alone rises by one cedi on each of 20,000 dollars, some GHS 20,000, and the dollar-quoted freight and fuel surcharge rise proportionally with the same movement. The locally quoted cedi lines, such as terminal handling and inland transport, do not move. The build made at order time therefore no longer describes the cost, which is why the lesson's disciplines are to date every conversion, to re-run the build when the rate moves, and to budget with a stated planning rate, all before the selling prices derived from the old snapshot are relied on.

Answer 2 commits the snapshot error: the invoice's dollar amount is unchanged, but the cedi cost of those dollars is no longer the number in the build. Answer 3 inverts the exposure; currency movement touches the commercial lines first, while government charges follow their own legally defined bases on the day of assessment. Answer 4 reaches for the wrong instrument altogether: the customs value answers the legal question of what duty is computed on, and pricing from it understates the true cost exactly as Lesson 10.2's trainee error showed.

The takeaway: a build is alive only for the day of its rate; re-run it before each price review and each reorder.

Question 7 (tests lesson 10.3)

7

A consignee's container is discharged at Tema and made available for collection. For illustration, the booking allows five free days for collection, and the terminal's tariff grants four free storage days from the same availability. The container is collected on the ninth day after availability. Seven days after gate-out, the empty container is returned to the shipping line's depot, against a permitted return allowance of three days. In the following weeks the consignee receives invoices from both the shipping line and the terminal. All periods are illustrative for teaching purposes.

Which statement correctly accounts for the charges?

1. One of the two invoices must be erroneous, because a single container can incur only one time-based charge for any given day.

2. The terminal's charge is a duplicate of the line's demurrage, because storage and demurrage are two names for the same terminal-time charge.

3. The line bills demurrage for the four days the container stood in the terminal past its collection allowance, and detention for the four days the empty was retained past the return allowance; the terminal separately bills storage for the occupied days beyond its own shorter allowance, because demurrage and detention are the carrier's charges for its equipment and time while storage is the terminal's charge for its space. ✓

4. No charge can apply after collection, because collecting the container from the terminal ends every free-time clock on the shipment.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The journey runs through three charges with two creditors. Inside the terminal, the container stood four days past the carrier's five-day collection allowance, so the line bills demurrage for those chargeable days. The same visit also crossed the terminal's own shorter four-day storage allowance, so the terminal bills storage under its own tariff for the occupied days beyond it, a different organisation pricing a different resource, the space rather than the equipment's time inside the gate. After gate-out, the empty was retained seven days against a three-day return allowance, so the line bills detention for the four days its equipment was held outside the terminal. Demurrage and detention can legitimately arrive from the same creditor for different

legs, and storage from the second creditor alongside the first, which is why a delayed container can owe several charges for the same week.

Answer 1 is the double-billing suspicion the lesson corrects: the charges stack because they price different resources, and several invoices for one container can all be valid. Answer 2 merges the terminal's charge into the line's; storage is the facility's own tariff charge, not a second name for demurrage. Answer 4 assumes collection ends every clock; collection ends the demurrage leg and begins the detention leg, which ran until the empty's return.

The takeaway: place each time-based charge by its clock, where the container stood, under whose allowance, and whose tariff, and expect legitimate invoices to stack.

Question 8 (tests lesson 10.3)

8

An importer's booking at Tema carries, for illustration, a free-time allowance of six days and a progressive demurrage tariff: chargeable days one to five at GHS 250 per day, chargeable days six to ten at GHS 600 per day, and every chargeable day from the eleventh at GHS 1,000 per day. The container is collected on the tenth chargeable day. All figures are illustrative for teaching purposes.

What demurrage charge does the tariff produce?

1. GHS 4,250: five chargeable days at GHS 250 plus five chargeable days at GHS 600. ✓
2. GHS 2,500, because all ten chargeable days are priced at the first band's rate.
3. GHS 6,000, because all ten chargeable days are priced at the second band's rate.
4. GHS 3,650, because the day of collection is never billed, so only nine chargeable days are charged.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A progressive tariff prices each chargeable day from the band in which that day falls, and the reconstruction runs band by band. Chargeable days one to five sit in the first band: five days at GHS 250, which is GHS 1,250. Chargeable days six to ten sit in the second band: five days at GHS 600, which is GHS 3,000. The container was collected on the tenth chargeable day, so the third band is never reached, and the total is GHS 1,250 plus GHS 3,000 = GHS 4,250. Every chargeable day in the period is billed, including the day of collection, which a tariff's part-day convention commonly makes fully billable even when the truck arrives part-way through it.

Answer 2 applies the first band's rate to days that have already moved into the second band, and Answer 3 applies the second band to all ten days, including the five that belong to the first; both flatten a stepped structure into a single rate, and both change the total by real money. Answer 4 waives the collection day, which no tariff does: the chargeable period runs until collection, and dropping its day understates the charge by one full billable day. All four misreadings share one root, trusting a headline rate instead of rebuilding the charge from the tariff's bands, which is also how incorrect invoices slip past untrained eyes.

The takeaway: reconstruct every time-based charge band by band and day by day before accepting the figure, and audit the invoice against that reconstruction.

Question 9 (tests lesson 10.3)

9

An importer's consignment requires a regulatory permit that is still in process. Three working days before the last free day, the clearing agent reports that the approval will be late and that collection will overshoot the free-time allowance by about a week.

Which response reflects the discipline this module teaches?

1. Take no action yet: on a well-run shipment the expected time-based charges are zero and the shipment budget already says so, so it is too early to involve the shipping line.

2. Escalate immediately and ask the line, before the charges accrue, for an extension of free time, in writing, with the reason and a realistic new collection date; before accrual this is a routine commercial request, while the same request after accrual is an argument about money the tariff already says is owed. ✓

3. Wait for the demurrage invoice and then dispute it in writing, because charges caused by a public regulator's delay are automatically withdrawn on request.

4. Refuse any demurrage invoice in advance on the ground that free time cannot lawfully expire while a government permit is outstanding.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson's asymmetry is the whole answer: time-based charges are negotiable only before they accrue. With three working days before the last free day and a foreseeable overshoot of about a week, the escalation should leave the importer's office at once, as a written extension request to the line carrying the credible reason, a realistic new collection date, and the shipment's history of prompt collections behind it. Made then, it is the routine commercial

conversation that lines handle every day, and it can hold the exposure near zero. Made after accrual, the identical request becomes an argument about money the tariff already says is owed, conducted with a creditor that holds the records and owes the importer nothing.

Answer 1 twists the expected-zero budget into an excuse for passivity; the zero is a prediction with a process behind it, and the process at this moment is the early request, not silence. Answer 3 waits for the invoice and then assumes an automatic waiver for regulator-caused delay; no such rule exists, and disputes succeed on evidence about the charge itself, not sympathy about its cause. Answer 4 invents a legal suspension of free time that no tariff contains, and converts a solvable delay into a certain dispute that damages credibility.

The takeaway: the cheap day to solve a foreseeable delay is the day before the last free day; once the clock starts, only the invoice's price is left to argue about.

Question 10 (tests lesson 10.4)

10

A freight forwarder in Takoradi has cost-priced a project move at GHS 4,000. The commercial manager promises the board a profit margin of 20 per cent of the selling price. A colleague therefore quotes GHS 4,800, adding 20 per cent to the cost price. All figures are illustrative for teaching purposes.

Which option best evaluates the quotation?

1. The quotation is correct, because a 20 per cent markup on cost and a 20 per cent margin on the selling price are the same instruction expressed in different words.

2. The quotation overprices the job: a margin is already provided for inside a properly built cost price, so the correct quotation is GHS 4,000.

3. The quotation under-delivers the promise: GHS 4,800 carries only 800 GHS of margin, about 17 per cent of the selling price, while the promised margin requires a selling price of 4,000 divided by 0.80, which is GHS 5,000, on which 1,000 GHS is exactly 20 per cent. ✓

4. Either figure is acceptable, because markup and margin are two names for the same percentage and the board will read them interchangeably.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Margin and markup measure the same money on different bases: markup on the cost price, margin on the selling price. A promised margin of 20 per cent of the selling price means the

4,000-cedi cost price must form 80 per cent of the quotation, so the selling price is 4,000 divided by 0.80 = GHS 5,000, on which 1,000 GHS of margin is exactly 20 per cent. The colleague's GHS 4,800 adds a 20 per cent markup, 800 GHS, which leaves a margin of 800 out of 4,800, about 17 per cent of the selling price. The shortfall is small on one job and structural across a programme, because it repeats on every price built the same way, which is precisely how promised margins quietly fail to arrive.

Answer 1 states the confusion verbatim: the two percentages coincide only at zero, and nowhere else. Answer 2 denies that the quotation must carry the margin at all; the cost price is the floor below which the provider works at a loss, and the margin is added above it, never hidden inside it. Answer 4's equivalence is false for the same reason Answer 1 is, and a board promised a margin and shown a markup has been promised one number and shown another.

The takeaway: markup lives on the cost price, margin lives on the selling price, and a promise stated in one must be computed in the same one.

Question 11 (tests lesson 10.4)

11

An importer in Accra receives two quotations for the same machinery shipment from Shanghai, each totalling GHS 48,000. For illustration, the first covers port-to-port carriage to Tema only and is valid for 21 days; the second covers door-to-door delivery to the importer's Accra warehouse and states that it includes destination terminal handling, customs clearance fees, and inland delivery, and is valid for 14 days. The importer's decision will take about ten days. All figures are illustrative for teaching purposes.

Which option best describes the professional comparison?

1. The two are equally acceptable, because the totals are equal and both validity periods outlast the decision.

2. Equal totals do not describe equal purchases: the first offer omits destination terminal handling, clearance and inland legs that the second includes, so on a normalised door-to-door scope the first is effectively the dearer offer, and the second's stated inclusions should be confirmed in writing before acceptance. ✓

3. The first is automatically the better offer, because a port-to-port scope in which the buyer arranges destination services always costs less than having a forwarder include them.

4. The second's 14-day validity makes it the riskier offer regardless of scope, so the first should be chosen.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The evaluation checklist runs before the prices: normalise the scope, check the clock, test completeness. The clock is satisfied here, since both validity periods comfortably cover a ten-day decision. The scope is not: the first offer ends at Tema, leaving destination terminal handling, customs clearance fees and inland delivery to be arranged and paid separately, while the second claims those legs inside its total. Until the missing legs of the first offer are priced and added, the two 48,000-cedi figures describe different purchases, and on a normalised door-to-door basis the first is effectively the dearer offer by whatever its omitted legs cost. The second's claimed inclusions should then be converted from prose into confirmed written lines before acceptance, because silence and assertion are both tested in the same step.

Answer 1 compares headlines, and equal totals over unequal scopes are the classic case the normalisation step exists to catch. Answer 3 generalises from a habit to a rule; arranging destination services oneself is sometimes cheaper and sometimes dearer, and the word always is doing work that only pricing the missing legs can do. Answer 4 lets validity override scope; an offer that is live but narrow is the risky one, and a wide offer with adequate validity is not endangered by a longer validity elsewhere.

The takeaway: only offers normalised to the same scope, live for the decision window, and complete to the same visible content have prices that describe the same purchase.

Question 12 (tests lesson 10.4)

12

An exporter in Kumasi receives a logistics quotation for a consolidated shipment, valid for 30 days. Internal approvals delay the decision, and the exporter writes to accept the quoted figures on day 45. In the meantime, freight rates on the lane have fallen, and the forwarder's current input costs are now below the quoted price. All dates and figures are illustrative for teaching purposes.

Which statement best describes the position?

1. The exporter can insist on the original figures, because a quotation binds the provider until the customer chooses to release it.

2. The original price applies, because with the forwarder's costs lower, performing at the quoted price has become profitable, and profitability is what determines whether an offer can be enforced.

3. The acceptance relates back to the day of issue, so a binding agreement was formed on the day the quotation was written and its price governs.

4. The offer lapsed at the end of its validity whichever way the market moved; the forwarder responds promptly in writing, issues a fresh quotation at current inputs, and the price that governs is the one in the offer the parties actually accept, since honouring the original figures would be a commercial gesture rather than an obligation. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Quotation validity is the period during which the offer remains effective under its stated conditions, and it protects both sides by defining exactly how long the figures may be relied on. At day 45 the offer had been dead for a fortnight, and lapse does not ask which way the market moved: the offer's inputs changed and its window closed, so it simply ended. What follows is method, not grievance. The forwarder replies promptly and in writing, explains the position, and issues a fresh quotation at current inputs, here likely a lower one, and the price that governs is the one in the offer the parties actually accept. Honouring the original figures remains the forwarder's commercial choice, a gesture to a valued customer, never an obligation.

Answer 1 reverses the nature of the document: a quotation is an offer with its own terms, not a commitment that only the customer can release. Answer 2 makes profitability the source of obligation; profitability may make the forwarder willing, but only an offer alive within its window makes a price enforceable, and the two ideas must not be merged. Answer 3 invents a relation-back rule; acceptance binds an offer that is still alive, and cannot reach back past its own expiry to revive it.

The takeaway: match approval timelines to validity, or ask for an extension before expiry; where an offer lapses, the remedy is a fresh quotation openly accepted, in either market direction.